

PROVEIT RESEARCH CONTINUATION

Mixed Spectral Identities

Harmonic Newton Series, Dougall Jets,
Unequal Twists, and Stieltjes Collisions

Detailed proofs and reproducible computations

10 October 2026

Abstract. We continue four specific research directions in the ProveIt manuscript and its six incoming reports at the pinned repository snapshot. A normally convergent Newton expansion determines every spectral derivative of the entire harmonic remainder, including at nonpositive integers. Its shift jets are identified with classical multiple Arakawa–Kaneko functions, and a two-shift boundary yields generalized Stieltjes constants with finite repeated antiderivatives.

At the double residue zero of the centered Dougall family, we prove termwise polynomial collapse, a divided generating identity, and an all-order mixed-parameter table. Consequences include a weighted trigamma square and harmonic squares equal to $\zeta(2) \pm \zeta(3)$ and $3\zeta(4) - \zeta(5)$. An additional spectral variable generates every Stieltjes order. For unequal twists, we give a normally convergent Lerch expansion, all-index finite-part convolutions, exact finite reductions, and covariant antiderivatives.

For colliding Stieltjes products, we prove an all-factor resonant-subset completion theorem and exhibit a finite anomaly: the geometric triple collision with rates $(0, 1, 2)$ differs from the merged finite part by $\frac{1}{2}\zeta(2)\log 2$. We distinguish geometric collision limits, spectral restrictions, and iterated Laurent constants. The article includes detailed proofs, source corrections, reproducible diagnostics, and a focused research agenda. The retained Gaussian S_6 and S_8 conjectures are not resolved here.

Prepared with OpenAI for the ProveIt research programme.

Repository snapshot: `dcd95baeb7c0e914b138bddf6829c5b1d52b726`.

Ordinary mathematical proofs are distinguished from exact finite computations and numerical diagnostics. No proof-assistant formalization or arithmetic-independence claim is made.

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1 Scope, conventions, and source status

1.1 What this continuation establishes

The task is to extend the exact-identity programme in the ProveIt polylogarithm manuscript, not to reproduce results already supplied by its incoming research reports. We inspected the canonical manuscript and the six archives present in `docs/incoming` at commit `dcd95baeb7c0e914b138bdeef6829c5b1d52b726` [1, 2, 3, 4, 5, 6, 7]. The mathematical work concentrates on four gaps explicitly left by those sources. The following table records the resulting status.

Source target	Result and its precise scope
Higher spectral derivatives of the entire harmonic difference [5]	Theorem 2.3 and Corollary 2.4 give convergent formulas at every order. Theorem 2.5 identifies all shift jets with classical multiple Arakawa–Kaneko functions. A finite reduction of every general spectral derivative to ordinary zeta derivatives is not asserted.
Mixed Dougall jets at $b = c = 1$, Question 3 of [4]	Proposition 3.2 proves termwise collapse on the integer residue grid. Theorems 3.3 and 3.6 give every mixed parameter row and every mixed spectral–Stieltjes derivative.
Genuinely unequal twists, the first research question in [7]	Theorem 4.1 gives a convergent normal form at arbitrary orders. The all-index formula (4.18) and finite scalar reductions retain the phase and endpoint normalizations. The scalar classification is not an arithmetic independence theorem for Fourier samples.
Mixed triple collisions, Question 1 of [5]	Theorems 5.1 and 5.2 give an explicit triple collision expansion and the local completion for any finite number of factors. Unqualified equality of the geometric collision constant and the merged finite part is false. Theorem 5.6 also resolves the one-derivative cancellation target.

The results are proved below as analytic identities with specified domains and normalizations. Exact finite calculations are supporting certificates for selected algebraic steps. Numerical comparisons test implementations and signs; they do not supply the infinite identities or establish arithmetic independence. The supplied proofs have not been formalized in a proof assistant.

1.2 Classical antecedents and attribution

The basic functions and summations are established mathematics. We use the Hurwitz–zeta shift and derivative formulas, the polygamma series, the Lerch transcendent, Euler’s hypergeometric transformation, and the Rogers–Dougall ${}_5F_4(1)$ summation [11, 10, 13, 14, 8]. The multiple Arakawa–Kaneko functions were already defined in [15]; related multiple and Hurwitz extensions are treated in [16, 17]. The Hurwitz–Tornheim relation used in the collision section has the antecedent [18]. The centered resonance theorem and the entire harmonic subtraction are results of the supplied reports. Where needed, their analytic mechanisms are reproduced to make the new deductions reviewable without reconstructing an entire earlier archive.

The contribution claimed here is the collection of deductions, normal forms, convergence arguments, and corrections relative to that inspected source set. We make no exhaustive priority claim over the full special-functions literature. In particular, a recognition of a classical

Arakawa–Kaneko function is recorded as an identification, not as the invention of a new special function.

1.3 Notation and the meaning of finite part

Write $(a)_n = \Gamma(a+n)/\Gamma(a)$ for a rising factorial, $(a)^n = a(a-1)\cdots(a-n+1)$ for a falling factorial, and $[u^r]$ for coefficient extraction. The symbols $\{\frac{m}{j}\}$ and $[\frac{m}{j}]$ denote Stirling numbers of the second kind and unsigned Stirling numbers of the first kind, respectively. Empty products are 1. The ordinary generalized harmonic numbers are

$$H_n^{(r)} = \sum_{k=1}^n k^{-r}, \quad H_0^{(r)} = 0, \quad H_n = H_n^{(1)}.$$

General shifts are defined explicitly in the relevant sections. The letters a, b, u are local parameters and are reintroduced when their role changes.

We set $\psi = \Gamma'/\Gamma$, and $\psi^{(r)}$ denotes an argument derivative. For zeta functions, $\zeta^{(k)}(s, x)$ denotes a derivative in the first, spectral variable; derivatives in x are written ∂_x . The generalized Stieltjes constants use the convention

$$\zeta(1+z, x) = \frac{1}{z} + \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \gamma_m(x) z^m, \quad \gamma_m = \gamma_m(1), \quad \gamma = \gamma_0. \quad (1.1)$$

In particular $\gamma_0(x) = -\psi(x)$. Principal logarithms are used for powers in the right half-plane. The phases for two-sided Fourier powers are fixed separately in (4.2). A square at a complex shift is an algebraic square, not an absolute-value square.

A series is *normally convergent* on a parameter domain when it converges absolutely and uniformly on each compact subset. This is the hypothesis used for holomorphic differentiation of infinite sums. The distribution-valued version means normal convergence after pairing with each smooth test function, together with the displayed uniform growth estimates on Fourier coefficients.

For a cutoff expansion in a positive ambient distance η , FP retains the coefficient of $\eta^0(\log \eta)^0$ after the divergent local terms are removed. At each singular grid point in Section 5, the cutoff is on the right in the ambient coordinate x . This specifies a value; a change of coordinate can change it. The symbol CT_ϵ has the analogous meaning for a geometric collision parameter ϵ . For multivariate meromorphic functions with poles on sums of variables, an unqualified rectangular Laurent constant is not a definition. A chamber, a spectral restriction, or the holomorphic completion must first be specified.

1.4 Reading the four proof mechanisms

The harmonic argument first proves a compact-uniform estimate for finite differences, then identifies their Mellin transform. This order prevents the invalid differentiation of a formula known only at discrete spectral values. The Dougall argument keeps the Gamma ratios and the finite small-index terms intact until after the subtractions have been differentiated. The unequal-twist argument isolates finitely many modes before applying a binomial expansion, so that the logarithm branches agree on the remaining tail. The collision argument works with the complete local numerator of each resonant subset, retaining finite terms as well as residues. These are the analytic steps on which the extensions depend.

2 A Newton expansion for the entire harmonic difference

The exact-identities report [5] leaves open the explicit evaluation of the higher spectral derivatives of its entire common-shift remainder. Its finite Stirling formula has an infinite, normally convergent extension. This extension gives every spectral derivative, continues the shift farther as harmonic depth increases, and identifies all shift jets at the unshifted base point with multiple Arakawa–Kaneko functions. Those functions are classical; the contribution here is their precise connection with the repository’s entire difference and the convergent spectral-derivative formula. No literature-wide priority is asserted.

2.1 Definitions and the known entire remainder

For $\Re a > 0$, use principal powers and put

$$F_a(s, u) = \sum_{n \geq 0} \frac{(a+u)_n}{(a)_n(n+a)^s}, \quad K_a(u) = \frac{\Gamma(a)\Gamma(1+u)}{\Gamma(a+u)},$$

$$D(s, u; a) = F_a(s, u) - K_a(u)F_1(s, u).$$

The first series initially converges for $\Re s > 1 + \Re u$. The source proves that D is entire in s , locally near $u = 0$. If

$$e_r(n; a) = [u^r] \prod_{\nu=0}^{n-1} \left(1 + \frac{u}{a+\nu}\right), \quad M_r(s; a) = \sum_{n \geq 0} e_r(n; a)(n+a)^{-s},$$

then its depth coefficients are

$$D_r(s; a) = [u^r]D(s, u; a) = M_r(s; a) - \sum_{j=0}^r [u^j]K_a(u) M_{r-j}(s; 1).$$

In particular $D_0(s; a) = \zeta(s, a) - \zeta(s)$, with its removable singularity at $s = 1$ understood. Each D_r is entire in s .

We use the source’s exact Mellin remainder

$$R_a(t, u) = -\frac{a-1}{1+u} e^{-at} {}_2F_1(1, a+u; 2+u; 1-e^{-t}), \quad (2.1)$$

$$D(s, u; a) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} R_a(t, u) dt, \quad \Re s > 0. \quad (2.2)$$

To check the normalization directly, the geometric generating function is

$$\sum_{n \geq 0} \frac{(a+u)_n}{(a)_n} e^{-(n+a)t} = e^{-at} {}_2F_1(1, a+u; a; e^{-t}) = K_a(u) \frac{e^{-t}}{(1-e^{-t})^{1+u}} + R_a(t, u).$$

The Gauss connection formula about $z = 1$ gives the last equality: the regular coefficient is $-(a-1)/(1+u)$ and the singular coefficient is $K_a(u)$; the remaining singular Gauss factor reduces to $e^{(a-1)t}$ before multiplication by e^{-at} . This initially holds for generic parameters, then throughout the stated germ by continuation [14]. The kernel R_a is analytic at zero and decays exponentially at infinity. Taking Mellin transforms proves (2.2) first in the original common half-plane and then for $\Re s > 0$ by continuation. Taylor subtraction at zero proves the entire continuation in s .

Define the finite exponential polynomial

$$A_j(s) = \sum_{\ell=1}^j (-1)^{j-\ell} \binom{j}{\ell} \ell^{1-s}, \quad j \geq 1. \quad (2.3)$$

It is entire in s . At a nonpositive integer,

$$A_j(-m) = j! \binom{m+1}{j}, \quad m \geq 0, \quad (2.4)$$

and therefore it vanishes for $j > m + 1$.

2.2 Normal convergence, including at nonpositive integers

Lemma 2.1 (Finite-difference Mellin estimate). *For $\Re s > 1 - j$,*

$$A_j(s) = \frac{(-1)^{j-1} j}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-t} (1 - e^{-t})^{j-1} dt. \quad (2.5)$$

At zeros of $1/\Gamma(s)$ this is its analytic product with the convergent integral. For every compact $S \subset \mathbb{C}$ there are constants C, P such that

$$\sup_{s \in S} |A_j(s)| \leq C(1 + \log(j+1))^P \quad (j \geq 1). \quad (2.6)$$

Every fixed spectral derivative obeys a bound of the same form, with constants depending on its order and S .

Proof. For $\Re s > 0$, expand the finite binomial power and use $j \binom{j-1}{\ell-1} = \ell \binom{j}{\ell}$. Termwise Laplace integration gives (2.5). The integrand at zero is $O(t^{\Re s + j - 2})$, so its integral is holomorphic on $\Re s > 1 - j$. Analytic continuation gives the formula on that larger half-plane. Thus it holds near any prescribed nonpositive integer as soon as j is sufficiently large.

Write $\sigma_- = \min_{s \in S} \Re s$ and $\sigma_+ = \max_{s \in S} \Re s$. Choose an integer N such that $\sigma_- + N - 1 > 0$, and consider $j \geq N$. Put $q = 1 - e^{-1} < 1$. On $0 < t \leq 1$,

$$(1 - e^{-t})^{j-1} \leq t^{N-1} q^{j-N},$$

so the integral on this interval, including its factor j , is at most $jq^{j-N}/(\sigma_- + N - 1)$. This is exponentially small. On $t \geq 1$, put $P = \max(0, \sigma_+ - 1)$. The function

$$p_j(t) = je^{-t}(1 - e^{-t})^{j-1}$$

is a probability density on $(0, \infty)$. With $L_j = 1 + 2 \log(j+1)$, the integral on $[1, L_j]$ is bounded by L_j^P . On $[L_j, \infty)$ use $p_j(t) \leq je^{-t}$ and

$$j \int_{L_j}^\infty t^P e^{-t} dt \leq C_P j e^{-L_j} (1 + L_j)^P \leq C'_P (1 + \log(j+1))^P.$$

The reciprocal gamma function is bounded on S . The finitely many remaining entire functions A_j are harmless. Cauchy's estimate on a slightly enlarged compact proves the derivative assertion. $\square$

Lemma 2.2 (Uniform binomial estimate). *For a compact set $A \subset \mathbb{C}$,*

$$\left| \binom{a-1}{j} \right| \leq C_A j^{-\Re a} \quad (a \in A, j \geq 1).$$

Proof. Use $(-1)^j \binom{a-1}{j} = \prod_{\nu=1}^j (1 - a/\nu)$. For $\nu > 2 \max_{a \in A} |a|$, $\log |1 - a/\nu| = -\Re a/\nu + O_A(\nu^{-2})$, uniformly. Summing and bounding the initial finite factors proves the claim, including when a passes through a positive integer. $\square$

Theorem 2.3 (Normally convergent Newton expansion). *For $\Re a > 0$ and $u \notin \{-1, -2, \dots\}$,*

$$D(s, u; a) = - \sum_{j=1}^{\infty} \binom{a-1}{j} \frac{A_j(s)}{u+j}. \quad (2.7)$$

The series converges normally on compact subsets of the indicated domain with $s \in \mathbb{C}$. It extends the depth germ to a meromorphic function of u , with at most simple poles at negative integers, and

$$\operatorname{Res}_{u=-j} D(s, u; a) = - \binom{a-1}{j} A_j(s). \quad (2.8)$$

These poles are removable when the indicated residues vanish.

For every $r \geq 0$,

$$D_r(s; a) = (-1)^{r+1} \sum_{j=1}^{\infty} \binom{a-1}{j} \frac{A_j(s)}{j^{r+1}}. \quad (2.9)$$

This depth series converges normally on

$$s \in \mathbb{C}, \quad \Re a > -r, \quad (2.10)$$

and supplies a jointly holomorphic continuation there.

Proof. On a compact u -set avoiding the negative integers, $|u+j|^{-1} \leq C/j$. The lemmas bound the summands by $Cj^{-1-\delta}(1+\log(j+1))^P$, where $\delta = \min \Re a > 0$. In the depth series the same proof uses $\delta = \min(\Re a + r) > 0$. This proves normal convergence and permits arbitrary fixed derivatives in s and a .

To identify the sum, put $w = 1 - e^{-t}$. Euler's hypergeometric transformation [14] and then its series give

$$\begin{aligned} R_a(t, u) &= -\frac{a-1}{1+u} e^{-t} {}_2F_1(2-a, 1+u; 2+u; w) \\ &= e^{-t} \sum_{j=1}^{\infty} (-1)^j \binom{a-1}{j} \frac{j}{u+j} w^{j-1}. \end{aligned}$$

Initially u is near zero and $0 < w < 1$. For $\Re s > 0$ this series can be integrated absolutely against $t^{s-1} dt$: apply (2.5) with s replaced by $\Re s$, and the estimates already proved. Its Mellin transform is

$$\frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} R_a(t, u) dt = - \sum_{j \geq 1} \binom{a-1}{j} \frac{A_j(s)}{u+j}.$$

It equals D in a common domain; both sides are entire in s . The normal convergence gives the continuation and residues in u . Expand $(u+j)^{-1}$ on $|u| < 1$ and extract depth coefficients. The enlarged shift domain follows from direct normal convergence, not interpolation from integer shifts. $\square$

2.3 All spectral derivatives

Corollary 2.4 (Logarithmic finite-difference formula). *For $m, k, r \geq 0$,*

$$\partial_s^k D(-m, u; a) = (-1)^{k+1} \sum_{j \geq 1} \frac{\binom{a-1}{j}}{u+j} \sum_{\ell=1}^j (-1)^{j-\ell} \binom{j}{\ell} \ell^{m+1} (\log \ell)^k, \quad (2.11)$$

$$\partial_s^k D_r(-m; a) = (-1)^{r+k+1} \sum_{j \geq 1} \frac{\binom{a-1}{j}}{j^{r+1}} \sum_{\ell=1}^j (-1)^{j-\ell} \binom{j}{\ell} \ell^{m+1} (\log \ell)^k. \quad (2.12)$$

The first formula is normally convergent for $\Re a > 0$, u away from the negative integers; the second for $\Re a > -r$. At $k = 0$ use $(\log 1)^0 = 1$.

Proof. Differentiate the normally convergent series using the compact spectral-derivative estimates. $\square$

At $k = 0$ this is exactly the source's finite Stirling formula,

$$D(-m, u; a) = - \sum_{j=1}^{m+1} \left\{ \begin{matrix} m+1 \\ j \end{matrix} \right\} \frac{(a-1)^j}{u+j}.$$

For $k \geq 1$ the finite difference generally no longer vanishes at large j . The answer is an absolutely convergent infinite identity; no finite reduction to polygamma functions at arbitrary depth is claimed. A finite value formula valid only at discrete values of s cannot itself be differentiated in s .

For clarity, if $[\epsilon^k]$ denotes a Laurent coefficient at $s = -m$, the identity defining D_r gives, for $k \geq 0$,

$$\begin{aligned} [\epsilon^k] M_r(-m + \epsilon; a) &= \sum_{j=0}^r [u^j] K_a(u) [\epsilon^k] M_{r-j}(-m + \epsilon; 1) \\ &\quad + \frac{1}{k!} \partial_s^k D_r(-m; a). \end{aligned} \quad (2.13)$$

Thus the theorem determines every coefficient of the entire shift correction. It does not by itself reduce the unshifted height-one Laurent coefficients, or the entire outer-shift family of the earlier report, to a finite basis of ordinary zeta derivatives.

Depth-zero special cases give useful independent checks:

$$\log \Gamma(a) = \sum_{j \geq 1} \frac{\binom{a-1}{j}}{j} \sum_{\ell=1}^j (-1)^{j-\ell} \binom{j}{\ell} \ell \log \ell, \quad (2.14)$$

$$\gamma_k(a) - \gamma_k(1) = - \sum_{j \geq 1} \frac{\binom{a-1}{j}}{j} \sum_{\ell=1}^j (-1)^{j-\ell} \binom{j}{\ell} (\log \ell)^k. \quad (2.15)$$

The Stieltjes convention is $\zeta(1+z, a) = z^{-1} + \sum_{k \geq 0} (-1)^k \gamma_k(a) z^k / k!$. The first identity follows from Lerch's formula [11] and the second from the regular expansion of D_0 at $s = 1$. These are specializations of a Newton/Hasse type calculus, with no claim that the classical log-gamma or Stieltjes expansions are new.

There is a finite reduction at every nonnegative integer *depth-generating parameter* $u = q$. For $q \geq 1$ and initially $\Re a > 0$,

$$D(s, q; a) = \frac{1}{(a)_q} \sum_{\ell=1}^q \left[\begin{matrix} q \\ \ell \end{matrix} \right] \{ \zeta(s-\ell, a) - \zeta(s-\ell) \}, \quad (2.16)$$

where brackets denote unsigned Stirling numbers of the first kind. Indeed $(a+q)_n / (a)_n = (n+a)_q / (a)_q$, and expand the rising factorial polynomial for both F_a and $K_a F_1$. All spectral derivatives follow by replacing each zeta by its order derivative. In particular

$$\partial_s^k D(-m, 1; a) = \frac{\zeta^{(k)}(-m-1, a) - \zeta^{(k)}(-m-1)}{a}.$$

Every zeta difference in (2.16) is entire, including at its removable principal pole.

2.4 The Arakawa–Kaneko boundary and every shift jet

For a positive integer multi-index $\mathbf{k} = (k_1, \dots, k_p)$ use the descending-index convention

$$\mathrm{Li}_{\mathbf{k}}(z) = \sum_{n_1 > \dots > n_p \geq 1} \frac{z^{n_1}}{n_1^{k_1} \dots n_p^{k_p}}.$$

Define the classical multiple Arakawa–Kaneko function by

$$\Xi(\mathbf{k}; s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{e^t - 1} \mathrm{Li}_{\mathbf{k}}(1 - e^{-t}) dt, \quad \Re s > 0. \quad (2.17)$$

Write $\Xi_r(s) = \Xi((r); s)$ for $r \geq 1$. Arakawa–Kaneko [15] already define the multiple function. They use ascending summation indices, so $\Xi(r+1, \{1\}^{p-1}; s)$ here corresponds to their $\xi(1, \dots, 1, r+1; s)$; the vector is reversed. At depth one there is no reversal ambiguity, and

$$\Xi_1(s) = s\zeta(s+1),$$

with a removable singularity at $s = 0$.

For completeness all functions in (2.17) are entire in s . The polylogarithm is analytic at $z = 0$ and begins at degree p , so its kernel is analytic at $t = 0$, beginning at order $p - 1$. At infinity its growth before multiplication by $(e^t - 1)^{-1}$ is at most polynomial: since all indices are positive, its positive terms are bounded by those of $\mathrm{Li}_{1, \dots, 1}(1 - e^{-t}) = t^p/p!$. Taylor subtraction at zero, followed by the zeros of $1/\Gamma(s)$, therefore proves entireness. This is also the continuation in the classical theory [15, 16].

Theorem 2.5 (Boundary values and mixed shift jets). *For $r \geq 1$, the continuation in (2.10) gives*

$$D_r(s; 0) = (-1)^r \Xi_r(s). \quad (2.18)$$

For every $r \geq 0$ and $p \geq 1$,

$$\boxed{\partial_a^p D_r(s; a)|_{a=1} = (-1)^{r+p} p! \Xi(r+1, \underbrace{1, \dots, 1}_{p-1}; s).} \quad (2.19)$$

The identities hold for every $s \in \mathbb{C}$, and arbitrary fixed spectral derivatives can be applied to both sides. Consequently

$$D_r(s; a) = (-1)^r \sum_{p=1}^{\infty} (1-a)^p \Xi(r+1, \{1\}^{p-1}; s), \quad |a-1| < r+1. \quad (2.20)$$

The series is locally normal jointly with s .

Proof. For $a = 0$, $\binom{-1}{j} = (-1)^j$. Substitute into the normally convergent depth series and apply (2.5):

$$D_r(s; 0) = \frac{(-1)^r}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-t} \sum_{j \geq 1} \frac{(1 - e^{-t})^{j-1}}{j^r} dt.$$

For $\Re s > 0$, termwise integration is justified absolutely. The sum is $\mathrm{Li}_r(w)/w$, where $w = 1 - e^{-t}$, giving (2.18). Continue in s .

For the shift jets let $e_j(N)$ be the elementary symmetric polynomial of degree j in $1, 1/2, \dots, 1/N$, with $e_0(N) = 1$ and $e_j(N) = 0$ for $j > N$. The finite product

$$\binom{a-1}{n} = \frac{(-1)^{n-1}(a-1)}{n} \prod_{\nu=1}^{n-1} \left(1 - \frac{a-1}{\nu}\right)$$

gives

$$\partial_a^p \binom{a-1}{n} \Big|_{a=1} = \frac{(-1)^{n-p} p!}{n} e_{p-1}(n-1).$$

Normal convergence permits insertion into (2.9). The Mellin integral then contains

$$\sum_{n \geq 1} \frac{w^{n-1}}{n^{r+1}} e_{p-1}(n-1) = \frac{1}{w} \text{Li}_{r+1, \{1\}^{p-1}}(w),$$

with overall sign $(-1)^{r+p} p!$. Absolute integration follows from positive coefficients and the logarithmic growth $e_{p-1}(n-1) = O((1 + \log n)^{p-1})$. This proves (2.19) on $\Re s > 0$, then everywhere.

Finally $D_r(s; 1) = 0$, and D_r is holomorphic for $\Re a > -r$. Its Taylor series centered at 1 therefore converges throughout $|a - 1| < r + 1$. Cauchy estimates on compact s -sets prove joint local normality. $\square$

The depth-zero case of (2.19) reduces to

$$\Xi(\underbrace{1, \dots, 1}_p; s) = \frac{(s)_p}{p!} \zeta(s + p).$$

The boundary formula (2.18), however, excludes $r = 0$. The original domain does not include $a = 0$, and

$$D_0(s; a) = a^{-s} + \zeta(s, a + 1) - \zeta(s)$$

has no jointly holomorphic extension there for general s . A fictitious Ξ_0 with a nondecaying Mellin kernel cannot remove this obstruction.

Some exact examples, with $\gamma_k = \gamma_k(1)$, are

$$\begin{aligned} D_1(s; 0) &= -s\zeta(s + 1), \\ D_1(0; 0) &= -1, & \partial_s D_1(0; 0) &= -\gamma, & \partial_s^2 D_1(0; 0) &= 2\gamma_1, \\ \partial_s^k D_1(0; 0) &= (-1)^k k \gamma_{k-1} \quad (k \geq 1), \\ \partial_s D_1(-1; 0) &= \frac{1}{2} - \frac{1}{2} \log(2\pi), \\ \partial_s D_1(-2; 0) &= \frac{1}{12} + 2\zeta'(-1), \\ \partial_a D_1(s; 1) &= \Xi_2(s), & \partial_a^2 D_1(s; 1) &= -2\Xi(2, 1; s). \end{aligned}$$

They are identities for the continued remainder, rather than ordinary sums of its divergent defining series.

More generally, for an integer $N \geq 0$ and depth $r \geq N + 1$,

$$D_r(s; -N) = \frac{(-1)^r}{N!} \sum_{h=0}^N \begin{bmatrix} N+1 \\ h+1 \end{bmatrix} \Xi_{r-h}(s). \quad (2.21)$$

Indeed $\binom{-N-1}{j} = (-1)^j \binom{N+j}{N}$ and

$$\binom{N+j}{N} = \frac{1}{N!} \prod_{\nu=1}^N (j + \nu) = \frac{1}{N!} \sum_{h=0}^N \begin{bmatrix} N+1 \\ h+1 \end{bmatrix} j^h.$$

Insert this polynomial into the convergent Mellin series. The condition $r > N$ makes all polylogarithm indices positive and places the shift inside $\Re a > -r$. For example

$$D_2(s; -1) = \Xi_2(s) + s\zeta(s + 1).$$

Specializing the convergent Taylor expansion to $a = 0$ also gives $\Xi_r(s) = \sum_{p \geq 1} \Xi(r + 1, \{1\}^{p-1}; s)$ for $r \geq 1$, normally on compact s -sets. Its elementary polylogarithmic explanation is $\sum_{p \geq 1} e_{p-1}(n-1) = n$. This is recorded as a consequence of the shift identification, with no new-priority assertion for the height-one summation itself.

2.5 A boundary corollary for generalized Stieltjes constants

The two-shift bridge in the source admits the same proof. Write $v = b - a$, $c = 1 + v$, and initially assume $\Re a, \Re b, \Re c > 0$. Define

$$D_{a,b}(s, u) = \sum_{n \geq 0} \frac{(a+u)_n}{(a)_n(n+b)^s} - K_a(u) \sum_{n \geq 0} \frac{(1+u)_n}{n!(n+c)^s}, \quad D_{a,b,r}(s) = [u^r] D_{a,b}(s, u).$$

Its entire Mellin remainder is $e^{-vt} R_a(t, u)$. Put

$$A_j(s; v) = \sum_{\ell=1}^j (-1)^{j-\ell} \binom{j}{\ell} \ell(\ell+v)^{-s}, \quad \Re v > -1.$$

The Laplace calculation gives

$$A_j(s; v) = \frac{(-1)^{j-1} j}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-(1+v)t} (1 - e^{-t})^{j-1} dt, \quad \Re s > 1 - j.$$

Then

$$D_{a,b,r}(s) = (-1)^{r+1} \sum_{j \geq 1} \binom{a-1}{j} \frac{A_j(s; b-a)}{j^{r+1}}, \quad (2.22)$$

normally on the connected domain

$$\Re(1 + b - a) > 0, \quad \Re b > -r, \quad s \in \mathbb{C}. \quad (2.23)$$

This supplies a continuation without a restriction on $\Re a$.

Here is the convergence detail needed for this enlarged domain. On compact s, v sets with $\Re v > -1$, and $0 < \epsilon < \min \Re(1 + v)$ sufficiently small, the modified Mellin estimate is $|A_j(s; v)| \leq C_\epsilon j^{-\Re v + \epsilon}$. On $(0, 1)$ the previous proof is unchanged; on $t \geq 1$ absorb the necessary power of t into $C_\epsilon e^{\epsilon t}$ and use

$$j \int_0^\infty e^{-(1+\Re v - \epsilon)t} (1 - e^{-t})^{j-1} dt = j B(1 + \Re v - \epsilon, j) \leq C_\epsilon j^{-\Re v + \epsilon}.$$

The last bound follows directly by bounding $(1 - x)^{j-1}$ by $e^{-(j-1)x}$ in the beta integral. Since $a + v = b$, choose ϵ below half of $\min(\Re b + r)$ on the compact under consideration. The series is then bounded by a summable power $C j^{-1 - (\Re b + r) + \epsilon}$. Identification uses the same kernel expansion as before. All fixed spectral derivatives are allowed by normal convergence.

For $r \geq 1$, $\Re x > 0$, define the classical Hurwitz extension

$$\Xi_r(s, x) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{e^{-xt}}{1 - e^{-t}} t^{s-1} \text{Li}_r(1 - e^{-t}) dt.$$

This convention agrees with Coppo–Candelpergher [17], and $\Xi_r(s, 1) = \Xi_r(s)$. It is entire in s by the same Taylor subtraction argument. The point $a = 0$, $b = x - 1$ lies in (2.23) for $r \geq 1$, so

$$D_{0,x-1,r}(s) = (-1)^r \Xi_r(s, x). \quad (2.24)$$

At depth one, $\text{Li}_1(1 - e^{-t}) = t$ gives

$$D_{0,x-1,1}(s) = -s \zeta(s+1, x) = \partial_x \zeta(s, x). \quad (2.25)$$

The right side is entire in s , including at zero.

Corollary 2.6 (Generalized Stieltjes boundary jets). *For $\Re x > 0$ and $k \geq 1$,*

$$\boxed{\partial_s^k D_{0,x-1,1}(s) \Big|_{s=0} = (-1)^k k \gamma_{k-1}(x).} \quad (2.26)$$

The zeroth derivative is -1 . For $m \geq 1$,

$$\partial_s^k D_{0,x-1,1}(s) \Big|_{s=-m} = m \zeta^{(k)}(1-m, x) - k \zeta^{(k-1)}(1-m, x),$$

with the last term omitted when $k = 0$. At $k = 1, m = 0$ the boundary function is $\psi(x)$, and its antiderivative is $\log \Gamma(x)$.

Proof. Insert the defining Laurent expansion of the Hurwitz zeta function in (2.25). The factor s removes its pole before coefficient extraction. $\square$

Every one of these boundary jets also has a finite repeated antiderivative. For $m, k \geq 0$ and $q \geq 1$, define coefficients by

$$\prod_{\nu=1}^{q-1} \left(1 - \frac{z}{m+\nu}\right)^{-1} = \sum_{h \geq 0} h_h^{[m, q-1]} z^h$$

(the product is 1 at $q = 1$). Then

$$\mathcal{P}_{m,k}^{(q)}(x) = \frac{k!}{(m+1)_{q-1}} \sum_{j=0}^k \frac{h_{k-j}^{[m, q-1]}}{j!} \zeta^{(j)}(1-m-q, x) \quad (2.27)$$

satisfies

$$\frac{d^q}{dx^q} \mathcal{P}_{m,k}^{(q)}(x) = \partial_s^k D_{0,x-1,1}(s) \Big|_{s=-m}.$$

Indeed a q -fold primitive of $-s\zeta(s+1, x)$ is $\zeta(s-q+1, x)/(1-s)_{q-1}$. Its denominator is nonzero near $s = -m$, and differentiating k times in s proves the formula. The coefficients are complete homogeneous symmetric polynomials in $1/(m+1), \dots, 1/(m+q-1)$, equivalently Bell polynomials in generalized harmonic-number differences. Other q -fold primitives differ by a polynomial of degree less than q .

2.6 Scope and further questions

No substantive error was found in the inspected fully-shifted and outer-shift harmonic arguments. Their restriction of the finite coefficient rule to the principal part and finite part is correct. The new formulas identify the missing analytic information in higher spectral derivatives. Three extension requirements are explicit: a discrete finite value formula cannot be differentiated in s ; depth and spectral manipulations must use the already entire remainder instead of disregarding the moving polar divisors of F_a ; and the common-shift $a = 0$ Arakawa–Kaneko formula requires positive depth.

Further questions include identifying finite combinations of spectral jets that reduce to ordinary Hurwitz-zeta derivatives; determining the exact shift singularities on $\Re a = -r$; developing stable evaluation of the finite differences, since naive binomial summation loses precision; comparing the present shift deformation systematically with existing generalized Arakawa–Kaneko families; and seeking multiple-zeta or iterated-integral identities for mixed spectral and shift derivatives beyond the standard integer specializations.

3 Mixed Dougall derivatives at a double residue zero

The centered resonance theorem already evaluates the minimal subtraction at every nonpositive integer excess parameter. Its residue-zero loci contain further identities: after restriction to some loci the remainder vanishes term by term, while derivatives transverse to those loci produce nonzero sums. We make this statement precise, give a finite formula at every mixed order, and evaluate a weighted square of trigamma differences for an arbitrary positive shift. An additional spectral variable gives an all-order generating function for generalized Stieltjes constants.

The underlying summation is the classical Rogers–Dougall ${}_5F_4(1)$ formula, DLMF 16.4.9 [8]. The summation itself and the centered subtraction theorem are antecedents, not novelty claims [3, 4]. The formulas below resolve the explicit mixed-parameter target at $b = c = 1$ in the incoming coordinate-transport report. No claim of global literature priority is required for these deductions.

3.1 Exact polynomial collapse on the integer grid

Use the centered notation

$$\eta = a/2, \quad x_n = n + \eta, \quad d_0 = 1 + a - b - c,$$

and

$$r_n(u) = x_n \frac{\Gamma(n+a)\Gamma(n+b)\Gamma(n+c)\Gamma(n+d_0-u)}{\Gamma(n+1)\Gamma(n+1+a-b)\Gamma(n+1+a-c)\Gamma(n+b+c+u)}.$$

Reciprocal Gamma factors are interpreted as entire functions, including at their zeros. Put

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (a/2, b - a/2, c - a/2, d_0 - u - a/2),$$

and define finite polynomial data by

$$\begin{aligned} h_j(u) &= -\frac{\sum_{i=1}^4 B_{2j+1}(\alpha_i)}{j(2j+1)}, \quad j \geq 1, \\ C_0(u) &= 1, \quad jC_j(u) = \sum_{k=1}^j kh_k(u)C_{j-k}(u). \end{aligned} \tag{3.1}$$

Here $B_j(z)$ denotes the Bernoulli polynomial with $B_1(z) = z - 1/2$. These are exactly the centered asymptotic coefficients:

$$r_n(u) \sim x_n^{-1-2u} \sum_{j \geq 0} C_j(u) x_n^{-2j}, \quad C_0 = 1.$$

At $u = -N$, $N \geq 1$, define the normally convergent bracket

$$F_{N,n}(a, b, c) = r_n(-N) - \sum_{j=0}^N C_j(-N) x_n^{2N-1-2j}.$$

Lemma 3.1 (Centered summation and residue transport: antecedent). *Suppose $a, b, c > 0$ and $d_0 > 0$. For $K \geq 1$ and $-K < \Re u < d_0$, the normally convergent series*

$$\begin{aligned} R_K(u) &= \sum_{n \geq 0} \left\{ r_n(u) - \sum_{j=0}^{K-1} C_j(u) x_n^{-1-2u-2j} \right\} \\ &= G(u) - \sum_{j=0}^{K-1} C_j(u) \zeta(1+2u+2j, a/2), \\ G(u) &= \frac{\Gamma(b)\Gamma(c)\Gamma(d_0-u)\Gamma(u)}{2\Gamma(d_0)\Gamma(b+u)\Gamma(c+u)}. \end{aligned} \tag{3.2}$$

The combined right side is holomorphic throughout this strip. For $N \geq 0$, let

$$\rho_N = \frac{(-1)^N}{N!} (d_0)_N (1-b)_N (1-c)_N.$$

Then

$$\begin{aligned} C_N(-N) &= \rho_N, \\ R_{N+1}(-N) &= \rho_N \left\{ \psi(a/2) + \frac{\psi(N+1) - \psi(b) - \psi(c) - \psi(d_0 + N)}{2} \right\}. \end{aligned} \quad (3.3)$$

All fixed spectral and parameter derivatives may be taken in the single bracket in (3.2).

Proof. At $x_n = n + a/2$, the denominator shifts are $1 - \alpha_i$, so

$$r_n(u) = x_n \prod_{i=1}^4 \frac{\Gamma(x_n + \alpha_i)}{\Gamma(x_n + 1 - \alpha_i)}.$$

The finite Stirling–Bernoulli expansion [9] and Bernoulli reflection give, uniformly on compact parameter sets,

$$r_n(u) = x_n^{-1-2u} \left\{ \sum_{j=0}^{K-1} C_j(u) x_n^{-2j} + O(n^{-2K}) \right\}.$$

Only even inverse powers occur; exponentiating the finite logarithmic expansion gives (3.1). Thus the bracket is summable throughout $-K < \Re u < d_0$. Cauchy's formula on a slightly larger compact set proves the same for each derivative, with fixed logarithmic powers allowed. In $0 < \Re u < d_0$, the classical Rogers–Dougall formula [8] sums $r_n(u)$ to $G(u)$: the upper parameters are $a, 1 + a/2, b, c, d_0 - u$, the lower parameters are $a/2, 1 + a - b, 1 + a - c, b + c + u$, and the conventional excess is $2u$. Separate summation of the counterterms gives (3.2) there. Holomorphic continuation proves the full-strip identity and removability of its apparent poles.

At $u = -N$ its Gamma residue is $\rho_N/2$. The only polar Hurwitz counterterm has index N and residue $C_N(-N)/2$. Equality of residues gives $C_N(-N) = \rho_N$; since these are polynomial coefficients, this equality extends through every residue zero.

For the constant term use

$$V = 2\partial_a + \partial_b + \partial_c - \partial_u, \quad W = \partial_a + \frac{1}{2}\partial_b + \frac{1}{2}\partial_c.$$

One has $V\alpha_1 = 1$ and $V\alpha_i = 0$ for $i = 2, 3, 4$. Consequently

$$Vh_k(u) = -\frac{B_{2k}(a/2)}{k} = 2\zeta(1-2k, a/2), \quad VC_j(u) = 2 \sum_{k=1}^j \zeta(1-2k, a/2) C_{j-k}(u).$$

The second formula follows by differentiating the formal exponential $\sum_j C_j y^j = \exp(\sum_k h_k y^k)$; every extracted coefficient is finite. At $u = -N$ it becomes

$$\frac{1}{2} C'_N(-N) + \sum_{j=0}^{N-1} C_j(-N) \zeta(1-2N+2j, a/2) = W\rho_N. \quad (3.4)$$

For generic b, c , the Gamma term has constant coefficient

$$\frac{\rho_N}{2} \{ \psi(N+1) - \psi(d_0 + N) - \psi(b - N) - \psi(c - N) \}.$$

The resonant Hurwitz term has constant coefficient $C'_N(-N)/2 - \rho_N\psi(a/2)$. Use (3.4) to combine this derivative with all lower counterterms. Since $Wd_0 = 0$,

$$W\rho_N = \frac{\rho_N}{2} \{\psi(b) - \psi(b-N) + \psi(c) - \psi(c-N)\}.$$

The shifted digammas cancel and yield (3.3). Its final arguments are positive, so normal convergence and holomorphy extend it across all zeros of ρ_N . These antecedent formulas reproduce the needed part of the earlier resonance construction [3, 4]. $\square$

Proposition 3.2 (Termwise collapse on every integer residue component). *Suppose $a, b, c > 0$ and $d_0 > 0$. For $1 \leq k \leq N$, at $c = k$,*

$$r_n(-N) = x_n \prod_{r=1}^{k-1} (x_n^2 - (\eta - r)^2) \prod_{r=0}^{N-k-1} (x_n^2 - (\eta + 1 - b + r)^2). \quad (3.5)$$

Consequently $F_{N,n}(a, b, k) = 0$ for every n , and the same statement holds with b and c interchanged. Locally along any intersection $b = h$, $c = k$ with $h, k \in \{1, \dots, N\}$, the holomorphic function $F_{N,n}$ is divisible by $(b-h)(c-k)$. Its quotient and every fixed derivative admit locally normally convergent sums. More globally, the quotient by $(1-b)_N(1-c)_N$ has a holomorphic extension throughout the stated parameter domain, and its sum is

$$\begin{aligned} & \sum_{n \geq 0} \frac{F_{N,n}(a, b, c)}{(1-b)_N(1-c)_N} \\ &= \frac{(-1)^N(d_0)_N}{N!} \left\{ \psi(a/2) + \frac{\psi(N+1) - \psi(b) - \psi(c) - \psi(d_0+N)}{2} \right\}. \end{aligned} \quad (3.6)$$

At a zero of either denominator factor, the summands in this formula mean their holomorphic extensions just described.

Proof. The Gamma functional equation gives

$$r_n(-N) = x_n(n+1)_{k-1}(n+1+a-k)_{k-1}(n+1+a-b)_{N-k}(n+b+k-N)_{N-k}.$$

Pair the factors symmetric about x_n to obtain (3.5). This is an odd polynomial of degree $2N-1$ in x_n . Uniqueness of asymptotic coefficients shows that its centered coefficients are exactly $C_0(-N), \dots, C_{N-1}(-N)$, whereas $C_N(-N) = 0$ and every subsequent coefficient is zero. The identity includes the finitely many small indices where an individual reciprocal Gamma factor vanishes.

For the divisibility statement, the holomorphic function vanishes when $b = h$ and when $c = k$. Successive one-variable Taylor expansions therefore factor $(b-h)(c-k)$. Cauchy's formula on a slightly larger parameter polydisc transfers the $O(n^{-3})$ uniform remainder bound to the divided function and to every prescribed derivative. This also proves normal convergence after division; a quotient cannot be justified merely by dividing two limiting sums at their common zero. Every zero of $(1-b)_N$ or $(1-c)_N$ is simple. The other local factors are units, so this establishes extension of the full divided sum. Its value follows from (3.3) off the zero loci and then by holomorphy on them. $\square$

The result is stronger than the vanishing of the residue polynomial or of the summed remainder. In particular, every derivative tangent to one of these components still has zero remainder termwise. Mixed transverse derivatives are the first potentially nonzero ones. For example, write

$$B_N(a, b, c) = \psi(a/2) + \frac{\psi(N+1) - \psi(b) - \psi(c) - \psi(d_0+N)}{2}.$$

At $b = h$, $c = k$ with $1 \leq h, k \leq N$,

$$\sum_{n \geq 0} \partial_b \partial_c F_{N,n}(a, h, k) = \frac{(-1)^{N+h+k} (d_0)_N}{N!} (h-1)!(N-h)!(k-1)!(N-k)! \times B_N(a, h, k), \quad d_0 = 1 + a - h - k. \quad (3.7)$$

This uses $\partial_b(1-b)_N|_{b=h} = (-1)^h(h-1)!(N-h)!$ and keeps all finite-head terms.

3.2 A two-parameter generating identity for all positive shifts

Write $b = 1 + p$, $c = 1 + q$, $A = a - 1$, and keep $u = -1$. Put

$$Q_n(a; p, q) = \frac{\Gamma(n+a)\Gamma(n+1+p)\Gamma(n+1+q)\Gamma(n+a-p-q)}{\Gamma(n+1)\Gamma(n+a-p)\Gamma(n+a-q)\Gamma(n+1+p+q)}, \quad (3.8)$$

$$B_a(p, q) = \psi(a/2) + \frac{\psi(2) - \psi(1+p) - \psi(1+q) - \psi(a-p-q)}{2}. \quad (3.9)$$

Thus $Q_n(a; p, 0) = Q_n(a; 0, q) = 1$, and $r_n(-1) = x_n Q_n(a; p, q)$.

Theorem 3.3 (Double-zero generating function). *For every $a > 0$ and sufficiently small complex p, q ,*

$$\sum_{n=0}^{\infty} \left\{ x_n(Q_n(a; p, q) - 1) + \frac{(A-p-q)pq}{x_n} \right\} = -(A-p-q)pq B_a(p, q). \quad (3.10)$$

The series is locally normally convergent in (a, p, q) in a neighborhood of each point $(a, 0, 0)$ with $\Re a > 0$. It may be differentiated at every fixed mixed order. Both sides are divisible by pq , and the divided identity also holds normally at $p = q = 0$.

Proof. For $a > 1$ and small p, q , Lemma 3.1 applies with $N = 1$. Its residue coefficient is

$$C_1(-1) = -(a-1-p-q)pq,$$

and the closed resonance value is exactly the right side of (3.10). The centered expansion supplies

$$Q_n(a; p, q) = 1 - \frac{(A-p-q)pq}{x_n^2} + O(n^{-4})$$

locally uniformly, so the displayed bracket is $O(n^{-3})$. All Gamma arguments in (3.8) lie in a pole-free neighborhood when $\Re a > 0$ and p, q are sufficiently small; the same centered expansion is uniform on compact subsets there. Thus the sum and the right side are holomorphic across $a = 1$ and throughout this connected positive-half-plane germ. The identity theorem extends their equality from $a > 1$ to all $a > 0$. For any prescribed compact a -set one can choose a smaller common polydisc in p, q and apply Cauchy's coefficient formula to the uniform remainder. This proves the derivative and division claims. $\square$

The restriction $d_0 > 0$ was needed only in the original region from which continuation was proved. After the exact cancellation above, $a = 1$ is a regular point, not an excluded parameter or a formal limit.

3.3 A finite Bell formula at every mixed order

For $k \geq 2$, define

$$\Lambda_k(n; a) = (-1)^k \psi^{(k-1)}(n+a) - \psi^{(k-1)}(n+1).$$

Taylor expansion of the four Gamma ratios gives the exact germ

$$\log Q_n(a; p, q) = \sum_{r,s \geq 1} \Lambda_{r+s}(n; a) \frac{p^r q^s}{r!s!}. \quad (3.11)$$

Define a finite bivariate Bell polynomial by

$$\mathcal{B}_{rs}(n; a) = r!s! [p^r q^s] \exp \left(\sum_{i,j \geq 1} \Lambda_{i+j}(n; a) \frac{p^i q^j}{i!j!} \right).$$

Only $i \leq r$, $j \leq s$, and powers at most $\min(r, s)$ of the exponent are needed. In particular,

$$\begin{aligned} \mathcal{B}_{11} &= \Lambda_2, & \mathcal{B}_{21} &= \Lambda_3, & \mathcal{B}_{31} &= \Lambda_4, \\ \mathcal{B}_{22} &= \Lambda_4 + 2\Lambda_2^2, & \mathcal{B}_{32} &= \Lambda_5 + 6\Lambda_2\Lambda_3, \\ \mathcal{B}_{33} &= \Lambda_6 + 9\Lambda_2\Lambda_4 + 9\Lambda_3^2 + 6\Lambda_2^3. \end{aligned}$$

Set $B_{rs} = \partial_p^r \partial_q^s B_a(0, 0)$, with

$$B_{00} = \psi(a/2) + \frac{1 + \gamma - \psi(a)}{2},$$

and, for $r + s = k \geq 1$,

$$B_{rs} = -\frac{1}{2} \left\{ \mathbf{1}_{s=0} \psi^{(r)}(1) + \mathbf{1}_{r=0} \psi^{(s)}(1) + (-1)^k \psi^{(k)}(a) \right\}. \quad (3.12)$$

Terms carrying a negative subscript below are omitted.

Corollary 3.4 (Complete mixed parameter table). *For $r, s \geq 1$, the single bracketed series*

$$S_{rs}(a) = \sum_{n \geq 0} \left\{ x_n \mathcal{B}_{rs}(n; a) + \frac{A \mathbf{1}_{r=s=1} - 2 \mathbf{1}_{(r,s)=(2,1)} - 2 \mathbf{1}_{(r,s)=(1,2)}}{x_n} \right\} \quad (3.13)$$

converges absolutely, and

$$S_{rs}(a) = -rs \{ A B_{r-1,s-1} - (r-1) B_{r-2,s-1} - (s-1) B_{r-1,s-2} \}. \quad (3.14)$$

For every $r + s \geq 4$, no counterterm is needed: the original polygamma polynomial $x_n \mathcal{B}_{rs}(n; a)$ is summable by itself.

Proof. Differentiate (3.10) and use (3.11). The counterterm is a polynomial of total degree three in p, q , which explains its exact disappearance at total derivative order four. Leibniz's rule gives (3.14); normal convergence was proved before differentiation, including the small indices. $\square$

For example, at $a = 2$ the first rows are

(r, s)	$\mathcal{B}_{rs}$	$S_{rs}(2)$
(1, 1)	Λ_2	0
(2, 1)	Λ_3	1
(3, 1)	Λ_4	$-6\zeta(3)$
(2, 2)	$\Lambda_4 + 2\Lambda_2^2$	$-4\zeta(3)$
(3, 2)	$\Lambda_5 + 6\Lambda_2\Lambda_3$	$24\zeta(3) - 18\zeta(4)$
(3, 3)	$\Lambda_6 + 9\Lambda_2\Lambda_4 + 9\Lambda_3^2 + 6\Lambda_2^3$	$108(\zeta(4) - \zeta(5))$.

The first two rows use the counterterms in (3.13). Every later row is an ordinary convergent sum without subtraction.

3.4 A closed square of trigamma differences

Theorem 3.5 (Weighted trigamma square). *For every real $a > 0$,*

$$\begin{aligned} & \sum_{n=0}^{\infty} \left(n + \frac{a}{2}\right) (\psi'(n+a) - \psi'(n+1))^2 \\ &= \frac{a-1}{4} \psi''(a) + \frac{\psi'(a) - \zeta(2)}{2} + \frac{3(a-1)}{2} \zeta(3). \end{aligned} \quad (3.15)$$

The series converges locally normally for $\Re a > 0$, with an ordinary algebraic square in that complex extension.

Proof. The (2, 2) and (3, 1) rows give respectively

$$\begin{aligned} \sum_n x_n (\Lambda_4 + 2\Lambda_2^2) &= 2A\psi''(a) + 4\psi'(a) - 4\zeta(2), \\ \sum_n x_n \Lambda_4 &= \frac{3A}{2} \{\psi''(1) + \psi''(a)\} + 3\{\psi'(a) - \zeta(2)\}. \end{aligned}$$

Subtract and divide by two, using $\psi''(1) = -2\zeta(3)$. Every series used in the subtraction is absolutely convergent. Independently, the polygamma asymptotic or its convergent series shows $\Lambda_2(n; a) = O(n^{-2})$ uniformly on compact subsets of $\Re a > 0$. Thus the weighted square is $O(n^{-3})$; this also directly verifies the domain and regularity at $a = 1$. $\square$

At $a = 3/2$, the functional equation and duplication formula [10] give

$$\psi'(3/2) = 3\zeta(2) - 4, \quad \psi''(3/2) = -14\zeta(3) + 16.$$

The right side of (3.15) becomes

$$\frac{1}{8}(-14\zeta(3) + 16) + \frac{1}{2}(2\zeta(2) - 4) + \frac{3}{4}\zeta(3) = \zeta(2) - \zeta(3).$$

For $H_n^{(r)} = \sum_{k=1}^n k^{-r}$, $H_0^{(r)} = 0$,

$$\psi'(n+1) - \psi'(n+3/2) = 2\{2H_{2n+1}^{(2)} - H_n^{(2)} - \zeta(2)\}.$$

Therefore

$$\boxed{\sum_{n=0}^{\infty} (4n+3) \{2H_{2n+1}^{(2)} - H_n^{(2)} - \zeta(2)\}^2 = \zeta(2) - \zeta(3).} \quad (3.16)$$

No regularized interpretation is involved in this positive series.

The continuation to $a = 1/2$ gives a companion identity,

$$\sum_{n=0}^{\infty} (4n+1) \{\zeta(2) - 2H_{2n}^{(2)} + H_n^{(2)}\}^2 = \zeta(2) + \zeta(3). \quad (3.17)$$

Here $\psi'(1/2) = 3\zeta(2)$ and $\psi''(1/2) = -14\zeta(3)$. At $a = 2$, the square formula reduces to $\sum_{k \geq 1} k^{-3} = \zeta(3)$, a useful elementary normalization.

The (3, 3) row also supplies a higher harmonic example. At $a = 2$, writing $k = n + 1$,

$$\Lambda_2 = -k^{-2}, \quad \Lambda_4 = -6k^{-4}, \quad \Lambda_6 = -120k^{-6}, \quad \Lambda_3 = 2\{2(\zeta(3) - H_{k-1}^{(3)}) - k^{-3}\}.$$

Substitution gives

$$\boxed{\sum_{k=1}^{\infty} k \{2(\zeta(3) - H_{k-1}^{(3)}) - k^{-3}\}^2 = 3\zeta(4) - \zeta(5).} \quad (3.18)$$

3.5 Every resonant polynomial moment from the same germ

For $N \geq 1$, $s = p + q$, set

$$P_{N-1}(x; s) = \prod_{r=0}^{N-2} \{x^2 - (a/2 - s + r)^2\}, \quad P_0 = 1.$$

The exact Gamma functional equation gives

$$r_n(-N) = x_n P_{N-1}(x_n; p + q) Q_n(a; p, q). \quad (3.19)$$

It retains every finite-head zero as a factor before derivatives are taken. This identity and the two-variable Bell germ generate the summands at every resonance without singular logarithmic derivatives.

More explicitly, write

$$P_{N-1}(x; s) = \sum_{k=0}^{N-1} P_k(s) x^{2N-2-2k}, \quad Q_n(a; p, q) \sim \sum_{j \geq 0} D_j(p, q) x_n^{-2j}.$$

Here $D_0 = 1$ and every D_j , $j \geq 1$, is divisible by pq . The counterterms are the finite convolution

$$C_j(-N) = \sum_{k=0}^{\min(j, N-1)} P_k(p + q) D_{j-k}(p, q). \quad (3.20)$$

Equation (3.3) becomes

$$\begin{aligned} \sum_{n \geq 0} F_{N,n}(a, 1 + p, 1 + q) &= \frac{(-1)^N}{N!} (a - 1 - p - q)_N (-p)_N (-q)_N \\ &\quad \times B_N(a, 1 + p, 1 + q). \end{aligned} \quad (3.21)$$

This holds as a normally convergent germ for every $a > 0$, by the same continuation argument as Theorem 3.3. Taking any prescribed $[p^r q^s]$ is a finite exact algorithm involving only the coefficients in (3.20), polygamma values, and finite generalized harmonic numbers from the Pochhammer factors. Formula (3.21) is a representation theorem; it does not assert that all possible products of polygamma functions can be separated individually by these mixed derivatives.

3.6 A mixed spectral germ containing every Stieltjes order

We now keep one derivative in each parameter direction and permit the spectral parameter to move: $u = -1 + t$. For $a > 0$, let

$$\begin{aligned} R_n(a; t) &= x_n \frac{\Gamma(n+1)\Gamma(n+a-t)}{\Gamma(n+a)\Gamma(n+1+t)}, \\ L_n(a; t) &= \psi(n+1) - \psi(n+a-t) + \psi(n+a) - \psi(n+1+t), \\ K_n(a; t) &= \psi'(n+a-t) - \psi'(n+1+t). \end{aligned} \quad (3.22)$$

At $p = q = 0$, these satisfy

$$\partial_p \partial_q r_n(-1 + t) = R_n(a; t) \{L_n(a; t)^2 + K_n(a; t)\}.$$

The identity uses the unspecialized summand before derivatives.

To state the endpoint without a division by $a - 1$, define

$$\begin{aligned} F_a(t) &= \frac{\Gamma(a-t)}{\Gamma(a)\Gamma(1+t)(1-t)}, \\ V_a(t) &= \psi(1) - \psi(a-t) + \psi(a) - \psi(1+t), \\ W_a(t) &= \psi'(a-t) - \psi'(a), \\ D_a(t) &= F_a(t) \left\{ 2t(1+tV_a(t)) - (a-1)[(1+tV_a(t))^2 + t^2W_a(t)] \right\}. \end{aligned} \quad (3.23)$$

All of these are holomorphic near $t = 0$ for every $a > 0$, including $a = 1$, and $D_a(0) = -(a-1)$.

Theorem 3.6 (Complete mixed spectral–Stieltjes germ). *For $a > 0$ and sufficiently small complex t , the series*

$$M_a(t) = \sum_{n=0}^{\infty} \left\{ R_n(a; t) (L_n(a; t)^2 + K_n(a; t)) + (a-1-2t)x_n^{-1-2t} \right\} \quad (3.24)$$

converges locally normally and equals

$$M_a(t) = \frac{D_a(t)}{2t} + (a-1-2t)\zeta(1+2t, a/2). \quad (3.25)$$

The apparent singularity at $t = 0$ is removable. Define $\gamma_m(v)$ by the Laurent expansion at the simple Hurwitz pole [11]:

$$\zeta(1+z, v) = z^{-1} + \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(v) z^m.$$

Then every fixed derivative has the explicit expression

$$M_a^{(m)}(0) = \frac{D_a^{(m+1)}(0)}{2(m+1)} - \mathbf{1}_{m=0} + (-2)^m \{ (a-1)\gamma_m(a/2) + m\gamma_{m-1}(a/2) \}. \quad (3.26)$$

The $m\gamma_{m-1}$ term is omitted when $m = 0$. Every derivative on the left can be taken in the single convergent bracket (3.24).

Proof. For $a > 1$, differentiate (3.2) with $K = 2$ once in p and once in q . The lower counterterm $C_0 = 1$ is parameter independent. Direct differentiation of the cubic Bernoulli coefficient gives

$$\partial_p \partial_q C_1(-1+t)|_{p=q=0} = -(a-1) + 2t.$$

The Gamma term is $E_1(t; p, q)/(2t)$, where

$$E_1(t; p, q) = \frac{\Gamma(1+p)\Gamma(1+q)\Gamma(a-p-q-t)\Gamma(1+t)}{\Gamma(a-1-p-q)\Gamma(1+p+t)\Gamma(1+q+t)} \frac{(p+t)(q+t)}{t-1}.$$

The factors $(p+t)(q+t)$ must be retained: setting $p = q = 0$ before differentiating would erase the mixed derivative. Twice applying the product rule and using

$$\Gamma(a) = (a-1)\Gamma(a-1), \quad \psi(a-1) = \psi(a) - (a-1)^{-1}, \quad \psi'(a-1) = \psi'(a) + (a-1)^{-2}$$

shows that $\partial_p \partial_q E_1(t; 0, 0) = D_a(t)$. The resulting expression (3.23) contains no singular factor at $a = 1$. This proves (3.25) for $a > 1$. Uniform Stirling remainders on a slightly enlarged parameter neighborhood give a bound $O(n^{-3-2\Re t}(1 + \log^j n))$ for every fixed t -derivative of the bracket. The displayed functions are holomorphic for $\Re a > 0$ near $t = 0$, so continuation gives all $a > 0$ as before.

Finally subtract the pole explicitly:

$$M_a(t) = \frac{D_a(t) + (a-1) - 2t}{2t} + (a-1-2t) \sum_{h \geq 0} \frac{(-2)^h}{h!} \gamma_h(a/2) t^h.$$

Taylor coefficient extraction yields (3.26), including the constant -1 at $m = 0$. The formula is for derivatives rather than divided Taylor coefficients; its factor $1/(m+1)$ is therefore essential. $\square$

For an explicit first spectral derivative, write

$$\ell = 1 + \gamma - \psi(a), \quad d = \psi'(a) - \zeta(2), \quad \mu_n = -\psi(n+a) - \psi(n+1).$$

Then (3.26) gives the ordinary identity

$$\begin{aligned} & \sum_{n=0}^{\infty} \left\{ x_n [\Lambda_3(n; a) + \mu_n \Lambda_2(n; a)] - \frac{2 + 2(a-1) \log x_n}{x_n} \right\} \\ &= 2\psi(a/2) + \ell - \frac{a-1}{4} (\ell^2 + 5d + 1) - 2(a-1) \gamma_1(a/2). \end{aligned} \quad (3.27)$$

Indeed the coefficient of t^2 in $D_a(t)$ is $2\ell - (a-1)(\ell^2 + 5d + 1)/2$. At $a = 2$, the right side reduces to $1 - \gamma^2 - 2\gamma_1$. Writing $k = n + 1$, this is equivalently

$$\sum_{k=1}^{\infty} \left\{ 4k(\zeta(3) - H_{k-1}^{(3)}) - \frac{1}{k^2} - \frac{2}{k} + \frac{2(H_{k-1} - \gamma - \log k)}{k} \right\} = 1 - \gamma^2 - 2\gamma_1. \quad (3.28)$$

The divergent-looking pieces in this last formula are one bracket; the proof supplies absolute convergence of their combination.

3.7 Further questions suggested by the mixed normal form

The arbitrary-order formulas above solve the requested first two-parameter table at $b = c = 1$ and provide its spectral extension. Several sharper questions remain. First, at a fixed homogeneous order, which individual polygamma-product moments are recoverable from the finitely many mixed Bell combinations? Their number need not match the number of all possible products, so the all-order generating theorem alone does not prove complete separation. Second, the polynomial formula on $b = h$, $c = k$ supplies starting points at every residue-grid intersection; compact explicit tables away from $(1, 1)$ should retain their finite-head zeros and avoid singular logarithmic derivative notation. Third, at rational a , the Stieltjes germ can be expressed in finite residue-class or Dirichlet-character coordinates; it is natural to determine which sectors cancel in particular mixed orders. Finally, primitive or generating-variable versions of the mixed identities can be tested for reductions to colored polylogarithms, with the function class and endpoint normalization stated before any claimed reduction.

4 Unequal twists: an exact normal form and its finite reductions

The incoming twisted-Stieltjes report asks for a normal form when two frequency shifts are genuinely different [7, research question on unequal twists]. An equal-twist Gamma quotient cannot by itself answer that question. We give three complementary answers: a convergent expansion in ordinary Lerch families, an all-index convolution formula with one bounded correction kernel, and a precise finite-reduction criterion for the underlying scalar functions. The last criterion concerns functional identities; it makes no inference about independence of sampled periods.

4.1 Fourier conventions and the entire family

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, $e_n(x) = e^{2\pi i n x}$, and $\widehat{U}(n) = \langle U, e_{-n} \rangle$. Let θ, η be real nonintegers, with $\theta \neq \eta$, and set

$$\lambda_{n,\theta} = 2\pi i(n + \theta), \quad (4.1)$$

$$L_{n,\theta} = \log(2\pi|n + \theta|) + \frac{\pi i}{2} \operatorname{sgn}(n + \theta). \quad (4.2)$$

Powers of $\lambda_{n,\theta}$ use this logarithm. The normalized single-twist family and the mixed family are defined by

$$\widehat{W}_s^\theta(n) = \lambda_{n,\theta}^{s-1}, \quad (4.3)$$

$$\widehat{\mathcal{K}}_{\alpha,\beta}^{\theta,\eta}(n) = \lambda_{n,\theta}^\alpha \lambda_{n,\eta}^\beta. \quad (4.4)$$

Both are entire with values in periodic distributions, jointly in all displayed spectral variables. Indeed their Fourier coefficients and every fixed number of order derivatives have polynomial growth, uniformly on compact spectral sets. Pairing with a smooth test function therefore gives a normally convergent series. In particular,

$$\mathcal{K}_{\alpha,\beta}^{\theta,\eta} = W_{1+\alpha}^\theta * W_{1+\beta}^\eta. \quad (4.5)$$

For $\Re(\alpha + \beta) < -1$, the Fourier series itself is absolutely and uniformly convergent. The distributional definition supplies all other spectral orders without assigning an ordinary value to a singular endpoint.

For $0 < \theta < 1$, the standard Lerch family gives, on $0 < x < 1$,

$$W_s^\theta(x) = \frac{e^{-2\pi i \theta x}}{\Gamma(1-s)} \Phi(e^{-2\pi i \theta}, s, x). \quad (4.6)$$

This identity is understood through analytic continuation when necessary. It follows initially by unfolding the Fourier coefficient of $\sum_{m \geq 0} e^{-(\varepsilon + 2\pi i \theta)(m+x)} (m+x)^{-s}$ and then letting $\varepsilon \downarrow 0$. This is also the normalization established in the incoming report. The Lerch function, its relation $\operatorname{Li}_s(z) = z\Phi(z, s, 1)$, and its analytic continuation are classical [13]. Other real twists follow by integer modulation, $W_s^{\theta+k} = e_{-k} W_s^\theta$.

4.2 A convergent Lerch expansion for arbitrary orders

Put $d = \theta - \eta$ and $h = 2\pi i d$. Choose a finite set $F \subset \mathbb{Z}$ such that, for some $\rho < 1$,

$$\left| \frac{d}{n + \eta} \right| \leq \rho \quad (n \notin F). \quad (4.7)$$

For example, all modes with $|n + \eta| \leq 2|d|$ may be put in F . Let $\Pi_F U = \sum_{n \in F} \widehat{U}(n) e_n$.

Theorem 4.1 (Normal convergence in single-twist Lerch coordinates). *For all $\alpha, \beta \in \mathbb{C}$,*

$$\begin{aligned} \mathcal{K}_{\alpha, \beta}^{\theta, \eta} &= \sum_{n \in F} \lambda_{n, \theta}^{\alpha} \lambda_{n, \eta}^{\beta} e_n \\ &+ \sum_{k=0}^{\infty} \binom{\alpha}{k} h^k (I - \Pi_F) W_{1+\alpha+\beta-k}^{\eta}. \end{aligned} \quad (4.8)$$

The series converges normally after pairing with any smooth test function, locally uniformly in (α, β) . Every fixed order derivative in either spectral variable may be taken termwise. After finitely many initial terms are removed, it converges normally in any prescribed C^q norm, locally on spectral parameter sets.

Proof. For $n \notin F$, condition (4.7) implies that $n + \theta$ and $n + \eta$ have the same sign. Therefore there is no branch correction in

$$L_{n, \theta} = L_{n, \eta} + \log \left(1 + \frac{d}{n + \eta} \right).$$

The ordinary binomial theorem proves the equality of every Fourier coefficient in (4.8). On a compact set of α , $|\binom{\alpha}{k}| \leq C(1+k)^M$ for suitable C, M . This elementary bound follows either from the finite product or by Cauchy's formula applied to $(1+z)^\alpha$ on a disc of radius between ρ and 1; the latter also gives a geometric bound sufficient here. Each summand is consequently bounded by a polynomial in $|n|$ times a summable geometric sequence in k . The test-function Fourier coefficients absorb the polynomial. Cauchy's formula on a slightly larger compact spectral set justifies all order derivatives.

For the C^q assertion, take k_0 larger than $q + 1 + \sup \Re(\alpha + \beta)$. Factor $|\lambda_{n, \eta}|^{\Re(\alpha + \beta) - k_0}$ from the terms with $k \geq k_0$. The remaining ratios are at most a fixed geometric sequence, while the q derivatives in x still leave an absolutely summable series over n . This proves the stated normal convergence. $\square$

The finitely many removed modes are essential when the twists lie in different integer intervals. Omitting them can change both the phase and the value of the result. The theorem gives a convergently evaluable normal form at every spectral order, including all Lerch order jets.

4.3 One bounded kernel controls every mixed Stieltjes jet

Let P_θ have Fourier coefficient $\gamma + L_{n, \theta}$, and define

$$B_\theta(u) = \Gamma(1-u) W_{1+u}^\theta = \delta_0 + \sum_{m \geq 0} \frac{(-1)^{m+1}}{m!} U_m^\theta u^{m+1}. \quad (4.9)$$

Thus $P_\theta = -U_0^\theta$. This agrees with the incoming finite-part Lerch normalization. Introduce the difference

$$A_{\theta, \eta} = P_\theta - P_\eta, \quad \widehat{A}_{\theta, \eta}(n) = L_{n, \theta} - L_{n, \eta}. \quad (4.10)$$

Lemma 4.2 (The difference is an ordinary bounded kernel). *If θ, η belong to the same open interval between consecutive integers, then*

$$A_{\theta, \eta}(x) = 2\pi i \int_\eta^\theta \frac{e^{-2\pi i t x}}{1 - e^{-2\pi i t}} dt, \quad 0 < x < 1. \quad (4.11)$$

This function extends smoothly to each closed side of the endpoint, is bounded, and has jump

$$A_{\theta, \eta}(0+) - A_{\theta, \eta}(1-) = 2\pi i(\theta - \eta). \quad (4.12)$$

For arbitrary real noninteger endpoints, the same formula holds with the path indented below the intervening integers. Equivalently, for $\eta < \theta$, it is the Cauchy principal value of the real integral plus $\pi i \sum_{k \in \mathbb{Z} \cap (\eta, \theta)} e_{-k}(x)$.

Proof. The Green kernel

$$G_t(x) = \frac{e^{-2\pi i t x}}{1 - e^{-2\pi i t}} \quad (4.13)$$

has Fourier coefficient $1/(2\pi i(n+t))$. Integrating that coefficient gives the logarithmic difference in (4.10) if no zero is crossed. Smoothness in x up to each side follows by differentiating the compact t -integral. Subtracting the endpoint values gives (4.12).

When t crosses $-n$, continuation below the pole adds $+\pi i$ to the logarithmic difference, precisely the principal-logarithm phase change from $-\pi/2$ to $+\pi/2$ in (4.2). The residue of $2\pi i G_t(x)$ at an integer k is $e_{-k}(x)$. Subtracting those finitely many simple poles defines an ordinary bounded function plus a finite Fourier polynomial. Its Fourier coefficients are again exactly (4.10). Reversing orientation covers $\theta < \eta$. $\square$

For $0 < \theta, \eta < 1$, an alternative useful expression is

$$A_{\theta, \eta}(x) = e^{-2\pi i \eta x} \Phi(e^{-2\pi i \eta}, 1, x) - e^{-2\pi i \theta x} \Phi(e^{-2\pi i \theta}, 1, x). \quad (4.14)$$

The two $1/x$ singularities cancel. If $\theta = p/q \in (0, 1)$, the corresponding term reduces to ordinary rational-shift digammas:

$$e^{-2\pi i p x/q} \Phi(e^{-2\pi i p/q}, 1, x) = -\frac{e^{-2\pi i p x/q}}{q} \sum_{j=0}^{q-1} e^{-2\pi i p j/q} \psi\left(\frac{x+j}{q}\right). \quad (4.15)$$

To prove it, split the Lerch defining sum into residue classes:

$$\Phi(e^{-2\pi i p/q}, s, x) = q^{-s} \sum_{j=0}^{q-1} e^{-2\pi i p j/q} \zeta\left(s, \frac{x+j}{q}\right).$$

Continue this identity to $s = 1$. The Hurwitz poles cancel because the finite character sum is zero; $\gamma_0(a) = -\psi(a)$ gives the formula.

Write $\exp_*(uA) = \delta_0 + \sum_{r \geq 1} u^r A^{*r}/r!$. Since A is bounded and belongs to $L^1(\mathbb{T})$, this exponential is entire with values in $\delta_0 + L^1(\mathbb{T})$, by the usual convolution norm estimate. Put

$$C(u, v) = \frac{\Gamma(1-u)\Gamma(1-v)}{\Gamma(1-u-v)}. \quad (4.16)$$

Theorem 4.3 (All-index unequal-twist formula). *As analytic germs of periodic distributions,*

$$B_\theta(u) * B_\eta(v) = C(u, v) B_\eta(u+v) * \exp_*(uA_{\theta, \eta}). \quad (4.17)$$

Consequently every mixed finite-part product is the finite coefficient formula

$$U_m^\theta * U_n^\eta = (-1)^{m+n} m! n! [u^{m+1} v^{n+1}] C(u, v) B_\eta(u+v) * \exp_*(uA_{\theta, \eta}). \quad (4.18)$$

*Only powers A^{*r} with $0 \leq r \leq m+1$ occur. The $r = 0$ part is the known equal-twist Gamma table; the other terms have Fourier coefficients tending to zero and introduce no new point-supported counterterm.*

Proof. At a Fourier mode, the right side of (4.17) is

$$\Gamma(1-u)\Gamma(1-v)e^{(u+v)L_{n,\eta}}e^{u(L_{n,\theta}-L_{n,\eta})},$$

which is exactly the left side. This proves the analytic identity by normal convergence after testing. Taking the indicated coefficients proves (4.18). For $r \geq 1$, the Fourier coefficient of A^{*r} is $O(|n|^{-r})$, whereas a fixed spectral jet of B_η grows only as a fixed polynomial in $\log |n|$. Thus their product tends to zero. A nonzero distribution supported at one point is a finite sum of derivatives of its Dirac mass and has polynomial Fourier coefficients; none can be added while preserving these coefficients. This statement refers to the displayed canonical decomposition, not to an arbitrary splitting of a distribution into summands. $\square$

The first symmetric identity is particularly compact:

$$U_0^\theta * U_0^\eta = U_1^\theta + U_1^\eta - \zeta(2)\delta_0 - \frac{1}{2}A_{\theta,\eta}^{*2}. \quad (4.19)$$

Indeed $2U_1^\theta = P_\theta^{*2} + \zeta(2)\delta_0$, and polarization gives the result. Here A^{*2} is an ordinary continuous function, since its Fourier coefficients are $O(n^{-2})$. The new term is therefore an explicitly evaluable ordinary convolution of bounded Lerch differences, while the contact coefficient remains fixed. An equal-twist formula with the correction omitted is false: at mode n it misses $\frac{1}{2}(L_{n,\theta} - L_{n,\eta})^2$.

4.4 Finite reductions and their precise boundary

Proposition 4.4 (Integer resolvent products). *For positive integers p, q , with $h = 2\pi i(\theta - \eta)$,*

$$\begin{aligned} \mathcal{K}_{-p,-q}^{\theta,\eta} &= \sum_{j=1}^p \frac{(-1)^q \binom{p+q-j-1}{p-j}}{h^{p+q-j}} W_{1-j}^\theta \\ &\quad + \sum_{j=1}^q \frac{(-1)^{q-j} \binom{p+q-j-1}{q-j}}{h^{p+q-j}} W_{1-j}^\eta. \end{aligned} \quad (4.20)$$

If one of α, β is a nonnegative integer, there is likewise a finite binomial reduction at the other twist. All integer pairs therefore reduce to finitely many single-twist families. In particular,

$$G_\theta * G_\eta = \frac{G_\eta - G_\theta}{2\pi i(\theta - \eta)}. \quad (4.21)$$

Proof. Expand $(A-h)^{-q}$ at $A=0$ and $(B+h)^{-p}$ at $B=0$, with $A-B=h$. The displayed coefficients are exactly the principal parts of $1/(A^p B^q)$ at its two poles. Their difference from that rational function is an entire rational function vanishing at infinity and is therefore zero. Substitute $A = \lambda_{n,\theta}$ and $B = \lambda_{n,\eta}$. The polynomial cases follow by the finite binomial theorem. Equation (4.21) is $p = q = 1$. $\square$

For positive integer j , the terms in (4.20) are elementary on the interval:

$$W_{1-j}^\theta(x) = \frac{(-1)^{j-1}}{(j-1)!(2\pi i)^{j-1}} \partial_\theta^{j-1} \frac{e^{-2\pi i \theta x}}{1 - e^{-2\pi i \theta}}. \quad (4.22)$$

The combined right side has its usual confluent limit W_{1-p-q}^θ as $\eta \rightarrow \theta$, although individual partial-fraction coefficients diverge. For example,

$$G_{1/4} * G_{3/4}(x) = \frac{i}{2\pi} \left((1+i)e^{-3\pi i x/2} - (1-i)e^{-\pi i x/2} \right), \quad (4.23)$$

$$G_{1/2}^{*2}(x) = e^{-\pi i x} \left(\frac{x}{2} - \frac{1}{4} \right). \quad (4.24)$$

Theorem 4.5 (Finite scalar reduction criterion). *Fix distinct complex numbers a, b and coherent branches on a simply connected domain avoiding $-a, -b$. Consider $f(z) = (z + a)^\alpha (z + b)^\beta$. It is a finite linear combination of single-centre powers and order jets*

$$(z + a)^r \text{Log}^k(z + a), \quad (z + b)^t \text{Log}^\ell(z + b), \quad k, \ell \in \mathbb{N} \cup \{0\},$$

with arbitrary complex exponents and constant coefficients, if and only if

$$\alpha \in \mathbb{Z}_{\geq 0}, \quad \text{or} \quad \beta \in \mathbb{Z}_{\geq 0}, \quad \text{or} \quad \alpha, \beta \in \mathbb{Z}. \quad (4.25)$$

Proof. Sufficiency follows from the binomial theorem and rational partial fractions. For necessity, suppose first $\alpha \notin \mathbb{Z}$ and $\beta \notin \mathbb{Z}_{\geq 0}$. Near $z = -a$, write $w = z + a$ and $c = b - a \neq 0$. Then

$$f(z) = w^\alpha \sum_{j \geq 0} \binom{\beta}{j} c^{\beta-j} w^j.$$

All infinitely many coefficients are nonzero. Terms based at $-b$ are holomorphic near $-a$. Apply analytic continuation once around $w = 0$ and subtract the original expression. On the left this multiplies the displayed infinite series by $e^{2\pi i \alpha} - 1 \neq 0$; on the proposed right it removes every term holomorphic there and leaves only finitely many powers times polynomials in $\text{Log } w$. Let $E = w d/dw$. A nonzero polynomial $P(E)$ annihilates that finite expression: for each occurring $w^r \text{Log}^k w$, include the factor $(E - r)^{k+1}$. Applied to the left, it gives a nonzero constant multiple of

$$w^\alpha \sum_{j \geq 0} P(\alpha + j) \binom{\beta}{j} c^{\beta-j} w^j.$$

Vanishing would force $P(\alpha + j) = 0$ for every $j \geq 0$, contrary to P being a nonzero polynomial. This is a contradiction.

Interchanging the two centres proves the same conclusion if $\beta \notin \mathbb{Z}$ and $\alpha \notin \mathbb{Z}_{\geq 0}$. Unless (4.25) holds, at least one of these two cases occurs. $\square$

The qualification “scalar” matters. Equality on the discrete set of Fourier modes is weaker than equality of the underlying holomorphic functions; the theorem does not promote the former to the latter. It explains exactly when the elementary functional mechanism behind (4.20) can terminate. Arbitrary orders still have the normally convergent expansion (4.8).

4.5 Argument derivatives and canonical primitives

Let $D_\theta = d/dx + 2\pi i \theta$. Fourier multiplication gives the exact contiguous and differential laws

$$D_\theta \mathcal{K}_{\alpha, \beta}^{\theta, \eta} = \mathcal{K}_{\alpha+1, \beta}^{\theta, \eta}, \quad D_\eta \mathcal{K}_{\alpha, \beta}^{\theta, \eta} = \mathcal{K}_{\alpha, \beta+1}^{\theta, \eta}, \quad (4.26)$$

$$\mathcal{K}_{\alpha+1, \beta}^{\theta, \eta} - \mathcal{K}_{\alpha, \beta+1}^{\theta, \eta} = h \mathcal{K}_{\alpha, \beta}^{\theta, \eta}. \quad (4.27)$$

Since neither twist is integral, D_θ and D_η are invertible on periodic distributions. Their inverses lower the corresponding spectral exponent. These statements remain valid after any order derivatives in α, β .

For an ordinary integrable f , the unique periodic covariant primitive has the explicit form

$$(D_\eta^{-1} f)(x) = e^{-2\pi i \eta x} \left\{ \int_0^x e^{2\pi i \eta t} f(t) dt + \frac{e^{-2\pi i \eta}}{1 - e^{-2\pi i \eta}} \int_0^1 e^{2\pi i \eta t} f(t) dt \right\}. \quad (4.28)$$

Differentiation proves $D_\eta U = f$, and the displayed constant is forced by $U(1) = U(0)$. This supplies ordinary quadratures for every bounded correction in (4.18), while the singular single-twist part retains its canonical distributional primitive.

5 Simultaneous Stieltjes collisions and their finite anomalies

The incoming separated-triple theorem assumes pairwise disjoint singular grids. Its first further question asks whether simultaneous collision constants agree with a directly merged ambient finite part [5, Question 1]. The bilinear coincidence report supplies such a comparison for two factors [6]. The triple problem contains an additional phenomenon: two singular factors can multiply a resonant Taylor coefficient of a third regular factor. We prove an explicit finite discrepancy, identify its dependence on the collision rates, and then give an all-factor local completion theorem. The distinction between cutoff and spectral prescriptions also appears in the different endpoint setting of [2]; here the collision parameter moves singularities in the real argument.

5.1 Fixed coordinates and the geometric collision theorem

Put $f(x) = \gamma_0(x) = -\psi(x)$ and $\gamma = \gamma_0(1)$. For $0 < a < b$ and $0 < \epsilon < 1/b$, let

$$C_{a,b}(\epsilon) = \text{FP}_x \int_0^1 f(x) f(\{x + a\epsilon\}) f(\{x + b\epsilon\}) dx.$$

At each of the three distinct grid points 0 , $1 - b\epsilon$, and $1 - a\epsilon$, the finite part deletes a right interval of ambient length η and takes the coefficient independent of η . The three cutoff variables can also be independent. Define the merged moment in the same coordinate by

$$Q = \text{FP}_x \int_0^1 f(x)^3 dx. \quad (5.1)$$

This has the ordinary convergent representation

$$Q = \int_0^1 \left[f(x)^3 - \frac{1}{x^3} - \frac{3\gamma}{x^2} - \frac{3(\gamma^2 - \zeta(2))}{x} \right] dx - \frac{1}{2} - 3\gamma. \quad (5.2)$$

Theorem 5.1 (Triple collision constant, with its rate dependence). *With $L = \log \epsilon$, define*

$$\begin{aligned} P_{-2}(a, b; L) &= \frac{L}{ab} + \frac{\log a}{a(b-a)} - \frac{\log b}{b(b-a)}, \\ P_{-1}(a, b; L) &= \gamma \left[\frac{L + \log a}{a} + \frac{L + \log b}{b} + \frac{L + \log(b-a)}{b-a} \right], \\ P_0(a, b; L) &= \zeta(2) \left[\frac{a-b}{a} (L + \log a) + \frac{b-a}{b} (L + \log b) + \frac{b}{b-a} (L + \log(b-a)) \right]. \end{aligned}$$

Then

$$C_{a,b}(\epsilon) = \epsilon^{-2} P_{-2}(a, b; L) + \epsilon^{-1} P_{-1}(a, b; L) + P_0(a, b; L) + Q + O(\epsilon |\log \epsilon|). \quad (5.3)$$

Consequently its coefficient independent of ϵ and $\log \epsilon$ is $Q + \zeta(2)D(a, b)$, where

$$D(a, b) = \frac{a-b}{a} \log a + \frac{b-a}{b} \log b + \frac{b}{b-a} \log(b-a).$$

In particular,

$$C_{1,2}(\epsilon) = \frac{L - \log 2}{2\epsilon^2} + \frac{\gamma(5L + \log 2)}{2\epsilon} + \frac{\zeta(2)}{2} (3L + \log 2) + Q + O(\epsilon |\log \epsilon|), \quad (5.4)$$

$$\text{CT}_\epsilon C_{1,2}(\epsilon) = Q + \frac{\zeta(2)}{2} \log 2 \neq Q. \quad (5.5)$$

Here CT_ϵ removes divergent powers and all nonconstant logarithmic terms; it does not remove finite constants inside a chosen collision counterterm. Removing the complete P_0 , including its finite part, does recover Q .

Proof. Choose a fixed small $\delta > 0$ and work in the circle coordinate $x \in (-\delta, \delta)$ around the merged point. The Hurwitz shift identity [11, Equation 25.11.3] gives the exact local representation

$$f(\{x\}) = \frac{\mathbf{1}_{x>0}}{x} + h(x), \quad h(x) = f(1+x) = \gamma - \zeta(2)x + x^2 R(x),$$

where R is analytic near zero. For the three rates $(0, a, b)$ write $S_c(x) = \mathbf{1}_{x+c\epsilon>0}/(x+c\epsilon)$ and $H_c(x) = h(x+c\epsilon)$. Expand the product of the three $S_c + H_c$. The all-regular term and the three terms with exactly one S_c are analytic in ϵ after the prescribed one-sided finite parts are taken. Their limits are their merged counterparts. The integral outside this neighborhood is also analytic for small ϵ .

For a term with two singular factors, let their rates be $c < d$ and let the remaining regular factor have rate e . Set $t = x + c\epsilon$, $\Delta = d - c$, $\rho = e - c$ and $\ell = \delta + c\epsilon$. Its local integral is

$$\text{FP} \int_0^\ell \frac{h(t + \rho\epsilon)}{t(t + \Delta\epsilon)} dt.$$

The two necessary elementary primitives are

$$\begin{aligned} A_0 &= \text{FP} \int_0^\ell \frac{dt}{t(t + \Delta\epsilon)} = \frac{\log \ell - \log(\ell + \Delta\epsilon) + \log(\Delta\epsilon)}{\Delta\epsilon}, \\ A_1 &= \int_0^\ell \frac{dt}{t + \Delta\epsilon} = \log(\ell + \Delta\epsilon) - \log(\Delta\epsilon). \end{aligned}$$

The constant and linear parts of h contribute $\gamma A_0 - \zeta(2)(A_1 + \rho\epsilon A_0)$. As $\epsilon \downarrow 0$, this equals

$$\frac{\gamma \log(\Delta\epsilon)}{\Delta\epsilon} + \zeta(2) \frac{d-e}{d-c} \log(\Delta\epsilon) - \frac{\gamma}{\delta} - \zeta(2) \log \delta + O(\epsilon |\log \epsilon|).$$

The last two displayed constants are exactly the merged finite parts of $\gamma x^{-2} - \zeta(2)x^{-1}$ on $(0, \delta)$. The analytic quadratic remainder converges to its merged ordinary integral. More precisely, use

$$\frac{(t + \rho\epsilon)^2}{t(t + \Delta\epsilon)} = 1 + \frac{(2\rho - \Delta)\epsilon}{t + \Delta\epsilon} + \frac{\rho^2\epsilon^2}{t(t + \Delta\epsilon)}$$

and the analytic, locally bounded function R to obtain an error $O(\epsilon |\log \epsilon|)$. Thus the three pairs give exactly $\epsilon^{-1}P_{-1} + P_0$ beyond their merged values.

The three-singular term is supported on $x > 0$. Partial fractions give

$$\frac{1}{x(x+a\epsilon)(x+b\epsilon)} = \frac{1}{abe^2x} - \frac{1}{a(b-a)\epsilon^2(x+a\epsilon)} + \frac{1}{b(b-a)\epsilon^2(x+b\epsilon)}.$$

Its lower-end contribution is exactly $\epsilon^{-2}P_{-2}$. The upper-end expression tends to $-1/(2\delta^2)$, which is precisely $\text{FP} \int_0^\delta x^{-3} dx$; its remaining error is $O(\epsilon)$. Summing all eight terms and the nonsingular exterior integral proves the formula. Only simple logarithms occur because the zero-index singular pieces are rational functions with simple linear factors; no $\log^2 \epsilon$ term has been suppressed. $\square$

The rate dependence can be isolated without an unevaluated merged constant:

$$\text{CT}_\epsilon C_{1,3}(\epsilon) - \text{CT}_\epsilon C_{1,2}(\epsilon) = \zeta(2) \left(\log 2 + \frac{2}{3} \log 3 \right). \quad (5.6)$$

These are positive shifts, exactly as in the definition of $C_{a,b}$. For three digamma factors all displayed correlations change sign.

5.2 Resonant subsets and a holomorphic completion

This section supplies an all-index explanation of the new phenomenon. Put

$$G(z, x) = \zeta(1 - z, x) + \frac{1}{z} = \sum_{m \geq 0} \gamma_m(x) \frac{z^m}{m!}, \quad c_r(z) = (-1)^r (1 - z)_r,$$

and write

$$g_r(z, x) = \partial_x^r G(z, x), \quad h_r(z) = g_r(z, 1).$$

In particular $h_0(z) = \zeta(1 - z) + 1/z$ and $h_r(z) = c_r(z)\zeta(r + 1 - z)$ for $r \geq 1$. All these functions are holomorphic near $z = 0$.

Let there be $d \geq 1$ factors, positive integer frequencies p_i , real shifts a_i , and nonnegative derivative orders r_i . Let $\mathcal{B}$ be the finite union of their singular grids on the circle, and define the collision set at $b \in \mathcal{B}$ by

$$C_b = \{i : p_i b + a_i \in \mathbb{Z}\}.$$

For $t = x - b > 0$ put

$$H_{b,j}(z_j, t) = \begin{cases} g_{r_j}(z_j, 1 + p_j t), & j \in C_b, \\ g_{r_j}(z_j, \{p_j b + a_j\} + p_j t), & j \notin C_b. \end{cases}$$

For every nonempty subset $S \subseteq C_b$ set

$$\Lambda_S = \sum_{i \in S} z_i, \quad k_S = \sum_{i \in S} (r_i + 1) - 1,$$

and define the complete resonant numerator

$$A_{b,S}(\mathbf{z}) = \left(\prod_{i \in S} c_{r_i}(z_i) p_i^{z_i - r_i - 1} \right) [t^{k_S}] \prod_{j \notin S} H_{b,j}(z_j, t).$$

Empty products are one. For $d \geq 2$, the term with $S = \{1, \dots, d\}$ vanishes because $k_S > 0$.

Theorem 5.2 (All-factor local completion). *Let $I_r(\mathbf{z})$ be the meromorphic continuation of the ordinary integral*

$$\int_0^1 \prod_{i=1}^d g_{r_i}(z_i, \{p_i x + a_i\}) dx,$$

initially defined, for example, by $\Re z_i > r_i + 1$ for every i . Then

$$J_r(\mathbf{z}) = I_r(\mathbf{z}) - \sum_{b \in \mathcal{B}} \sum_{\emptyset \neq S \subseteq C_b} \frac{A_{b,S}(\mathbf{z})}{\Lambda_S} \quad (5.7)$$

is holomorphic near the origin. Its coefficient

$$\left(\prod_i m_i! \right) [z_1^{m_1} \cdots z_d^{m_d}] J_r(\mathbf{z})$$

is exactly the ambient-coordinate finite part of the corresponding product of $\gamma_{m_i}^{(r_i)}$. The theorem permits arbitrary partial and complete collisions.

Proof. The exact Hurwitz shift gives each local factor as

$$g_{r_i}(z_i, \{p_i x + a_i\}) = c_{r_i}(z_i) p_i^{z_i - r_i - 1} t^{z_i - r_i - 1} + H_{b,i}(z_i, t) \quad (i \in C_b).$$

Expand the finite product by its nonempty subsets of singular pieces. For S , expand its regular complement through degree k_S . The Taylor coefficient of degree j integrates meromorphically with denominator $\Lambda_S + j - k_S$. All $j < k_S$ have denominator nonzero near the origin and already generate their coordinate finite parts. At $j = k_S$ the term is $A_{b,S} t^{\Lambda_S - 1}$. Its meromorphic integral on $(0, \ell)$ is $A_{b,S} \ell^{\Lambda_S} / \Lambda_S$, while its coefficientwise coordinate finite-part generator is $A_{b,S}(\ell^{\Lambda_S} - 1) / \Lambda_S$. Their difference is the complete numerator $A_{b,S} / \Lambda_S$, not a residue with its parameter dependence deleted. The Taylor remainder has exponent Λ_S and is integrable uniformly on a sufficiently small spectral polydisc; all fixed spectral derivatives add only logarithmic powers. Finite summation over grid points proves holomorphy and the coefficient interpretation, and also constructs the local meromorphic continuation uniquely from the initial convergent integral. $\square$

For two coincident factors, a product of both singular pieces has no regular complement of the positive degree needed for resonance. Hence the only polar divisors through the origin are $z_i = 0$. For three coincident factors, pairs of singular pieces have one regular complement, so $z_i + z_j = 0$ contributes for the first time. This is the structural difference behind the triple collision anomaly.

5.3 The explicit same-point triple generator

Corollary 5.3 (Every coincident triple Stieltjes jet). *At unit frequencies and a common endpoint, put $R = r_1 + r_2 + r_3$. The complete generator is*

$$\begin{aligned} J_{\mathbf{r}}(\mathbf{z}) = I_{\mathbf{r}}(\mathbf{z}) - \sum_{i=1}^3 \frac{c_{r_i}(z_i)}{z_i} \sum_{u+v=r_i} \frac{h_{r_j+u}(z_j) h_{r_k+v}(z_k)}{u!v!} \\ - \sum_{i<j} \frac{c_{r_i}(z_i) c_{r_j}(z_j) h_{R+1}(z_k)}{(r_i + r_j + 1)! (z_i + z_j)}. \end{aligned} \quad (5.8)$$

In each summand $\{i, j, k\} = \{1, 2, 3\}$. For $r_i = 0$ this is particularly short:

$$J_0 = I_0 - \sum_i \frac{h_0(z_j) h_0(z_k)}{z_i} - \sum_{i<j} \frac{h_1(z_k)}{z_i + z_j}. \quad (5.9)$$

Proof. For a singleton $S = \{i\}$, its resonant Taylor degree is r_i , and Leibniz's formula gives the inner sum in the first line. For a pair $S = \{i, j\}$, the resonant degree is $r_i + r_j + 1$. The remaining factor has derivative order r_k , so its Taylor coefficient is $h_{R+1}(z_k) / (r_i + r_j + 1)!$. The three-singular subset has no regular complement and makes no resonant contribution. Theorem 5.2 now gives both formulas. $\square$

The last sum is indispensable: $h_1(0) = -\zeta(2) \neq 0$.

5.3.1 A completely specified Tornheim realization

The classical Tornheim function is initially

$$T(A, B, C) = \sum_{m, n \geq 1} \frac{1}{m^A n^B (m+n)^C}.$$

Let $\alpha_i = z_i - r_i$ and define

$$K_3(\alpha) = \frac{2 \prod_i \Gamma(\alpha_i)}{(2\pi)^{\sum_i \alpha_i}} \sum_{k=1}^3 \cos\left(\frac{\pi}{2}(\alpha_i + \alpha_j - \alpha_k)\right) T(\alpha_i, \alpha_j, \alpha_k),$$

$$K_2(A, B) = \frac{2\Gamma(A)\Gamma(B)}{(2\pi)^{A+B}} \cos\left(\frac{\pi}{2}(A - B)\right) \zeta(A + B).$$

The classical Fourier calculation [18] gives these as the integrals of, respectively, three or two factors $\zeta(1 - \alpha, x)$ in an initial absolute-convergence domain. Its six sign cones explain the three cosines. For the regularized Stieltjes generators one has the exact meromorphic identity

$$I_r(z) = \left(\prod_i c_{r_i}(z_i) \right) K_3(\alpha) + \sum_{k:r_k=0} \frac{c_{r_i}(z_i)c_{r_j}(z_j)}{z_k} K_2(\alpha_i, \alpha_j) + \frac{\mathbf{1}_{r=0}}{z_1 z_2 z_3}.$$

Terms with two scalar $1/z_i$ factors vanish because the remaining Hurwitz factor has mean zero in the initial domain. This combined formula, followed by the complete local subtraction above, is a finite all-index coefficient rule. No arithmetic reduction of the Tornheim jets to single-zeta coordinates is asserted.

Meromorphic continuation of T near the required nonpositive integer orders can be proved directly from

$$T(A, B, C) = \frac{1}{\Gamma(C)} \int_0^\infty t^{C-1} \text{Li}_A(e^{-t}) \text{Li}_B(e^{-t}) dt.$$

On a small interval insert the classical local expansion [12, Equation 25.12.12]

$$\text{Li}_A(e^{-t}) = \Gamma(1 - A)t^{A-1} + \sum_{j \geq 0} \zeta(A - j) \frac{(-t)^j}{j!},$$

which is uniform on compact parameter neighborhoods avoiding positive integer A . Finite subtraction and termwise integration give explicit linear polar divisors; the remaining Mellin integral is holomorphic. The same argument handles the colored kernels with one or both twists equal to one. Thus an entire-order claim valid for two nontrivial twists must be weakened to meromorphic continuation when collisions force trivial twists. The incoming separated-grid theorem is valid on its stated hypotheses; it is its unrestricted extrapolation that would be incorrect.

5.4 Spectral rays and Laurent expansion chambers

Proposition 5.4 (Exact cubic spectral-ray correction). *Take $r_i = 0$ and $z_i = c_i t$, where $c_i \neq 0$ and $c_i + c_j \neq 0$. Since the completed J is holomorphic, its constant is always Q . The uncompleted meromorphic integral instead has constant*

$$\begin{aligned} \text{CT}_t I_0(c_1 t, c_2 t, c_3 t) &= Q + \gamma \gamma_1 \sum_i \frac{c_j + c_k}{c_i} \\ &\quad + (\zeta(2) + \zeta'(2)) \sum_{i < j} \frac{c_k}{c_i + c_j}. \end{aligned} \tag{5.10}$$

Here γ_1 is the ordinary first Stieltjes constant and ζ' is the spectral derivative. On the diagonal the discrepancy is

$$6\gamma\gamma_1 + \frac{3}{2}(\zeta(2) + \zeta'(2)).$$

Proof. Use (5.9) and the expansions $h_0(z) = \gamma + \gamma_1 z + O(z^2)$ and $h_1(z) = -\zeta(2) + (\zeta(2) + \zeta'(2))z + O(z^2)$. After division by $c_i t$ or $(c_i + c_j)t$, precisely the linear numerator coefficient contributes to the constant. The holomorphic remainder contributes $J_0(0, 0, 0) = Q$. For the diagonal, set $c_1 = c_2 = c_3 = 1$. $\square$

This operation concerns spectral rays and is distinct from the geometric collision $\epsilon \rightarrow 0$ proved earlier. Mixed poles also mean that a rectangular Laurent expansion is no longer a canonical operation unless an expansion chamber or an iterated prescription is specified. The holomorphic completion avoids either choice.

Proposition 5.5 (Three chamber constants for one derivative moment). *For $\mathbf{r} = (1, 0, 0)$ one has*

$$J_{1,0,0}(0, 0, 0) = \text{FP} \int_0^1 \gamma'_0(x) \gamma_0(x)^2 dx = 2\gamma\zeta(2) - \zeta(3). \quad (5.11)$$

In an iterated Laurent chamber ordering the three moduli, the raw constant is instead

$$\text{CT}_{\text{chamber}} I_{1,0,0} = \begin{cases} \zeta(3), & |z_1| \text{ is largest,} \\ 0, & |z_1| \text{ is intermediate,} \\ -\zeta(3), & |z_1| \text{ is smallest.} \end{cases} \quad (5.12)$$

On the spectral diagonal the same raw integral is identically zero: $I_{1,0,0}(t, t, t) = 0$ as a meromorphic function.

Proof. The local expansion is

$$\gamma_0(x) = x^{-1} + \gamma - \zeta(2)x + \zeta(3)x^2 + O(x^3).$$

Consequently the coefficient of x^0 in $\gamma_0(x)^3$ is $\gamma^3 - 6\gamma\zeta(2) + 3\zeta(3)$. Integrate the exact derivative $(\gamma_0^3)' / 3$ from the cutoff to 1 and retain its constant. This proves (5.11) directly, without a spectral regularization.

For clarity, write $I_{1,0,0} - J_{1,0,0} = S + P$, separating singleton and pair terms in (5.8). Since $c_1(z_1) = z_1 - 1$, these are

$$\begin{aligned} S &= \frac{z_1 - 1}{z_1} [h_1(z_2)h_0(z_3) + h_0(z_2)h_1(z_3)] \\ &\quad + \frac{h_1(z_1)h_0(z_3)}{z_2} + \frac{h_1(z_1)h_0(z_2)}{z_3}, \\ P &= \frac{(z_1 - 1)h_2(z_3)}{2(z_1 + z_2)} + \frac{(z_1 - 1)h_2(z_2)}{2(z_1 + z_3)} + \frac{h_2(z_1)}{z_2 + z_3}. \end{aligned}$$

The singleton constant is $-2\gamma\zeta(2)$ in every chamber. Using $h_2(0) = 2\zeta(3)$, the only pair terms contributing to a constant are

$$\zeta(3) \frac{z_1}{z_1 + z_2} + \zeta(3) \frac{z_1}{z_1 + z_3}.$$

In a chamber with $|z_j| < |z_1|$, expand $z_1/(z_1 + z_j) = \sum_{k \geq 0} (-z_j/z_1)^k$; its constant is one. In the reversed chamber its expansion starts with z_1/z_j and has constant zero. Thus P contributes $N\zeta(3)$, where N is the number of other variables dominated by z_1 . Adding the holomorphic constant in (5.11) gives $(N - 1)\zeta(3)$, proving all three cases.

Finally, for $\Re t > 2$, the function $G(t, x)$ has equal endpoint values and a continuous first derivative on the period. Therefore

$$I_{1,0,0}(t, t, t) = \int_0^1 \partial_x G(t, x) G(t, x)^2 dx = \frac{G(t, 1)^3 - G(t, 0)^3}{3} = 0.$$

Meromorphic continuation preserves this identity. In particular, its spectral diagonal constant is zero even though its coordinate finite part is (5.11). $\square$

In contrast, for $\mathbf{r} = \mathbf{0}$ and unit frequencies, the polar numerators do not depend on the variables in their own denominator. Their nonnegative rectangular coefficients vanish in every standard chamber. Thus the general mixed-pole warning should not be misread as asserting chamber dependence of the raw constant in that special zero-derivative case; its spectral-ray constants still differ, as shown above.

5.5 An exact cancellation at every number of factors

The incoming question on odd derivative order identifies equal Stieltjes indices and one total argument derivative as its first target [5, Question 3]. That case has a complete local answer. The mechanism is the elementary fundamental theorem of calculus, applied before taking the cutoff constant; keeping its endpoint term gives the following more general identity.

Theorem 5.6 (The all-factor total-derivative projection). *Let $d \geq 1$, $m_1, \dots, m_d \geq 0$, and put $Z_0 = \{i : m_i = 0\}$ and $g_i(t) = \gamma_{m_i}(1+t)$. Then*

$$\begin{aligned} \text{FP} \int_0^1 \sum_{i=1}^d \gamma'_{m_i}(x) \prod_{j \neq i} \gamma_{m_j}(x) dx \\ = - \sum_{\emptyset \neq S \subseteq Z_0} [t^{|S|}] \prod_{j \notin S} g_j(t). \end{aligned} \quad (5.13)$$

The sum on the right is empty when all $m_i > 0$. Each surviving coefficient is a finite expression in ordinary Stieltjes constants and argument derivatives of Stieltjes functions at 1.

Proof. For every fixed cutoff, the integrand is the ordinary derivative of $\prod_i \gamma_{m_i}(x)$. Hence its finite-part integral is

$$\prod_i \gamma_{m_i}(1) - \text{CT}_{x \downarrow 0} \prod_i \gamma_{m_i}(x).$$

Use the exact shift $\gamma_{m_i}(x) = x^{-1} \log^{m_i} x + g_i(x)$ and expand by the subset S of singular pieces. The empty subset contributes exactly $\prod_i \gamma_{m_i}(1)$, canceling the upper endpoint. A nonempty subset contributes $x^{-|S|} \log^{\sum_{i \in S} m_i} x \prod_{j \notin S} g_j(x)$. It has a constant in both x and $\log x$ precisely from the Taylor degree $|S|$ and only when $S \subseteq Z_0$. This proves (5.13). All individual finite parts exist by the same finite local power-log expansion, so taking the constant after the finite sum introduces no interchange issue. $\square$

For equal positive indices the theorem gives the requested cancellation at every number of factors:

$$\boxed{\text{FP} \int_0^1 \gamma'_m(x) \gamma_m(x)^{d-1} dx = 0, \quad d \geq 1, \quad m \geq 1.} \quad (5.14)$$

For the zero index, with $h(t) = \gamma_0(1+t)$ as above, the complete remaining local value is

$$\text{FP} \int_0^1 \gamma'_0(x) \gamma_0(x)^{d-1} dx = -\frac{1}{d} \sum_{k=1}^d \binom{d}{k} [t^k] h(t)^{d-k}. \quad (5.15)$$

This is a finite zeta polynomial because $h(t) = \gamma + \sum_{j \geq 1} (-1)^j \zeta(j+1) t^j$ near zero. For $d = 3$ it recovers (5.11); for $d = 4$ it gives

$$\text{FP} \int_0^1 \gamma'_0(x) \gamma_0(x)^3 dx = 3\gamma^2 \zeta(2) - 3\gamma \zeta(3) - \frac{11}{4} \zeta(4).$$

Thus the symmetric projection removes the multiple spectral terms entirely in this one-derivative case, leaving its explicitly specified local endpoint contribution. Higher odd derivative projections remain a separate question.

5.6 Repository scope and independent numerical diagnostics

The inspected snapshot is `dcd95baeb7c0e914b138bddef6829c5b1d52b726`. The incoming `Exact_Identities` report, section `triple_dilation.tex`, assumes pairwise disjoint grids throughout. Its `audit_research.tex`, Question 1, asks to remove that restriction and compare simultaneous collision constants with merged finite parts. The subset theorem completes its fixed-collision local subtraction question, and the explicit $(0, \epsilon, 2\epsilon)$ calculation shows that its suggested matching is not valid for plain geometric constant extraction. A finite rate-dependent correction is necessary. The preceding bilinear coincidence results are not contradicted.

The supplied script `code/verify_collisions.py` computes the separated moments by three independent convergent interval integrals. For an interval of length ℓ starting at the singular point of one factor, with the other two arguments $\eta_1 + t, \eta_2 + t$, it uses

$$\int_0^\ell \left[\frac{f(\eta_1 + t)f(\eta_2 + t) - f(\eta_1)f(\eta_2)}{t} + h(t)f(\eta_1 + t)f(\eta_2 + t) \right] dt + f(\eta_1)f(\eta_2) \log \ell.$$

This is exactly the ambient finite part with the simple pole visibly removed. It does not implement the collision expansion or the Tornheim representation. The merged moment is evaluated by a local digamma Laurent series plus ordinary quadrature, with two independent split points. At 70-digit working precision the computed values are

$$Q = -1.9662627343915343406643753384116314238258689764917 \dots,$$

and

$$\frac{1}{2}\zeta(2) \log 2 = 0.5700907053214263787372899294961746726083149206823 \dots$$

Numerical residuals tend to zero at the proved $O(\epsilon |\log \epsilon|)$ rate for $(a, b) = (1, 2), (1, 3), (2, 5)$. For the equally spaced example, the residual means $C_{1,2}(\epsilon) - \epsilon^{-2}P_{-2} - \epsilon^{-1}P_{-1} - P_0 - Q$; the complete finite term in P_0 is included in this diagnostic:

ϵ	Numerical residual
10^{-6}	$-9.332630963 \times 10^{-5}$
10^{-8}	$-1.237723998 \times 10^{-6}$
10^{-10}	$-1.542186195 \times 10^{-8}$

Omitting the finite term would instead leave a limit $\zeta(2) \log 2/2$. The full record contains seventeen collision diagnostics, the two-split merged calculation, working precision, and the finite correction. It is reproduced from the package root by

```
python code/verify_collisions.py --output results/collisions.json
```

These are floating-point diagnostics, not certified interval enclosures; the proof is the exact local decomposition above.

The classical inputs are the Hurwitz shift, argument-derivative and Fourier formulas [11], the local polylogarithm expansion [12], and the Hurwitz–Tornheim product relation [18]. The contributions here are the complete coordinate-normalized subtraction, the finite geometric collision anomaly, and their exact comparison with the specified spectral prescriptions. The statements do not claim a reduction of every Tornheim derivative to ordinary zeta values.

6 Source corrections, verification, and further research

6.1 Corrections proposed against the pinned manuscript

The inspection was targeted: the relevant harmonic, Dougall, twist, collision, and Gaussian-status material was checked, rather than every chapter and every archived computational receipt. The following two edits are proposed for `chapters/03-algebraic.tex` in the canonical manuscript [1]. They concern the exact snapshot specified in Section 1. The package contains a small unified patch; the repository itself has not been changed.

A real logarithm on the negative branch. In the equation labeled `cleo:eq:lintanh`, put $\theta = \arcsin \lambda$ for real $|\lambda| < 1$. The formula valid on both sides of zero is

$$\int_0^{\pi/2} \operatorname{arctanh}(\lambda \sin x) dx = \theta \log \left| \tan \frac{\theta}{2} \right| + 2 \operatorname{Cl}_2(\theta) - \frac{1}{2} \operatorname{Cl}_2(2\theta). \quad (6.1)$$

Here $\operatorname{Cl}_2(\theta) = \sum_{n \geq 1} \sin(n\theta)/n^2$. The value at $\theta = 0$ is understood by continuity. The original display omits the absolute value. For negative λ , the real logarithm of $\tan(\theta/2)$ is not defined, whereas the integral is real. Taking a principal complex logarithm would add an unwanted imaginary term. This is a carry-forward correction already identified in the incoming audit material, retained here because the pinned canonical display still needs it.

For a direct verification, differentiate the left side after $\lambda = \sin \theta$ to obtain $\theta/\sin \theta$. The derivative identity $\operatorname{Cl}_2'(\theta) = -\log |2 \sin(\theta/2)|$ shows that the three logarithmic terms in the derivative of the right side cancel, leaving the same $\theta/\sin \theta$. Both sides tend to zero at the origin. This proves (6.1) on the two intervals and through their continuous joining point.

Remove an unsupported non-elementarity assertion. The same chapter describes the $(1, 1)$ diagonal as the “simplest non-elementary diagonal.” Its exact evaluation

$$\int_0^{\pi/2} \arctan^2(\sin x) dx = \pi \chi_2(3 - 2\sqrt{2})$$

where $\chi_2(z) = \sum_{n \geq 0} z^{2n+1}/(2n+1)^2$, does not prove non-elementarity or minimality in any specified constant field. The chapter’s later paragraph correctly says that these properties remain unproved. We propose the neutral wording “For the $(1, 1)$ diagonal, the formula gives ...”. The value is unchanged. A finite unsuccessful relation search cannot justify a stronger assertion.

6.2 A corrected collision expectation, not a failed source theorem

The separated-grid theorem in [5] assumes disjoint singular supports and is not contradicted by this continuation. Nor is the proved bilinear coincidence comparison invalidated. What fails is an unqualified extension of the positive matching expectation in its Question 1 to geometric triple collisions. The exact counterexample is (5.5). A corrected research statement is:

For a specified collision path and ambient cutoff, determine the complete local collision expression, including its finite rate-dependent term. Subtract that expression before comparing the limit with the merged ambient finite part.

For the zero-index triple with arbitrary fixed rates $0 < a < b$, Theorem 5.1 proves precisely this statement, with the complete expression $\epsilon^{-2}P_{-2} + \epsilon^{-1}P_{-1} + P_0$. The all-factor theorem fixes the local meromorphic completion at a prescribed collision configuration; it does not by itself compute every geometric collision expansion for arbitrary Stieltjes indices and rates depending on different powers of ϵ . Those extensions are questions below.

6.3 Statements that must retain their current status

The source's S_2 and S_4 reductions already have proofs. The current S_6 candidate is labeled `cycloquot:conj:S6` in `chapters/04-cyclotomic-quotients.tex`; the retained S_8 candidate is labeled `s8new:conj:S8` in `chapters/04-S8-candidate.tex` [1]. Both remain conjectural here. The later S_8 vector must not be confused with the different older vector that the manuscript rigorously rejected. Exact equivalence of two candidate coordinate vectors proves equivalence, not vanishing of either residual. Likewise, an interval containing zero proves proximity only. No new Gaussian integer relation was fitted in this work.

The harmonic Newton theorem solves the explicit-representation problem for the entire remainder. It leaves the finite-basis reduction question open at general depth and spectral order. The Dougall Bell generator evaluates all its mixed combinations; it does not show that those combinations span every conceivable polygamma product. The unequal-twist scalar theorem concerns identities of functions with two distinct branch centers; it does not prove independence of their values on the integer Fourier lattice. These qualifications are part of the results.

6.4 Reproducibility and the role of computation

The analytic proofs supply the infinite summations, domains, continuations, and interchanges of limits. Four scripts implement independent checks or exact finite algebra. Their JSON records are included in `results/`; they contain the actual inputs, working precision, and residuals. The scripts need Python, `mpmath`, and `sympy`. The exact environment versions and commands are recorded in the package's `README.txt`.

Artifact	Checks and interpretation
<code>verify_harmonic.py</code>	Twenty-six numerical checks: twelve truncated Newton spectral derivatives against Hurwitz derivatives, one complex-parameter Mellin comparison, nine finite integer-shift specializations, and four repeated antiderivatives. The Newton series uses 240 terms and 180 working digits; its observed discrepancies are about 10^{-18} to 10^{-15} , set by truncation rather than the working precision.
<code>verify_dougall.py</code>	Thirteen numerical comparisons of the trigamma square, two-parameter Gamma identity, mixed spectral generator, and first spectral derivative. Six exact Bell-polynomial checks and twenty-one exact residue-grid factorizations use rational symbolic arithmetic. The numerical residuals are below 5.4×10^{-46} in the supplied run.
<code>verify_twists.py</code>	Forty-nine exact partial-fraction checks through $p, q = 7$, one deliberately corrupted-sign rejection, and the exact lowest-jet polarization identity. Eighteen numerical comparisons cover ordinary Green convolutions, confluent twists, the bounded logarithm difference, rational digamma conversion, branch crossings, Fourier coefficients, and complex-order tail modes.

Artifact	Checks and interpretation
<code>verify_collisions.py</code>	Seventeen separated-collision diagnostics at 70 working digits, plus a two-split evaluation of the merged integral. The independent interval quadratures visibly subtract each simple pole. Their residuals test the predicted asymptotic expansion; they are not exact-zero tests.

The finite symbolic checks do not formalize the analytic proofs. The floating-point comparisons are not interval enclosures. In particular, an estimated asymptotic tail term is not a certified error bound for the Gamma-ratio computations. The harmonic finite differences can lose many digits through cancellation; a larger working precision alone does not remove the separate Newton truncation error. These are concrete limitations of the supplied calculations, not qualifications of the proved normal-convergence theorems.

6.5 Further questions with concrete first targets

The following are proposals for additional exact research. They are not included in the proved theorem list.

Question 1 (Which individual polygamma moments can be separated?). At a fixed total order $w = r + s$, determine the exact span of the mixed Bell polynomials $\mathcal{B}_{rs}$, allowing the elementary linear polygamma rows and explicit parameter symmetries. The square identity used the difference of two rows at $w = 4$. At $w = 6$ and $w = 8$, give exact rational elimination matrices and identify which individual moments remain outside that span. An absence from this specified span should be reported as such, not as an impossibility of evaluation by other identities.

Question 2 (Compact formulas at every integer residue intersection). Use (3.7) to produce short explicit tables at $(b, c) = (1, 2)$ and $(2, 2)$ for $N = 2, 3$. Can central factorial polynomials organize the subtraction coefficients without expanding large Bernoulli polynomials? A useful certificate would retain the finite small-index zeros and replay the mixed derivative using only exact polynomial operations before evaluating polygamma endpoints.

Question 3 (Finite reductions of regular harmonic spectral jets). The Newton formula supplies an exact convergent representation; classify parameter configurations at which it reduces to finitely many ordinary Hurwitz-zeta order derivatives. Positive integral a and nonnegative integral u give starting cases. For positive depth and nonintegral rational a , first examine the first two derivatives at $s = 0, -1$. State the permitted function class before asking whether a multiple Arakawa–Kaneko order derivative is needed.

Question 4 (The exact boundary of shift continuation). The proved depth domain is $\Re a > -r$, and the Taylor series about $a = 1$ converges at least for $|a - 1| < r + 1$. Determine the actual singular germ at $a = -r$ for generic complex s and compare it with the exceptional nonpositive integral s , where the finite Stirling formula is polynomial in a . This should distinguish a sufficient convergence domain from a maximal analytic continuation and explain any exceptional cancellation of the first shift singularity.

Question 5 (Rational-shift character cancellations). For rational a , transform the full mixed spectral germ (3.26) into finite residue-class or Dirichlet-character coordinates. The first target is $a = 1/2$, where $\gamma_m(a/2) = \gamma_m(1/4)$ appears. At each prescribed order, compute an exact reduction matrix including every conductor logarithm. Which parameter symmetries cancel a whole character sector before numerical evaluation?

Question 6 (Unequal twists approaching an integer resonance). The finite-mode decomposition in (4.8) gives the exact modes that change phase. Let one or both twists approach an integer while the spectral orders are also differentiated. After isolating the resonant Fourier modes, determine a holomorphic completed germ and its endpoint contact terms. The first target is a simultaneous pair approaching zero at distinct fixed rates. The normalization should be compared with the already known single-twist degeneration.

Question 7 (From scalar functional reduction to sampled identities). The finite scalar criterion proved here uses local monodromy. Which additional hypotheses would let one infer a scalar identity from equality on every integer Fourier mode in the same finite power–logarithm class? An exact asymptotic uniqueness theorem at infinity is one possible route. Conversely, seek explicit sampled identities that do not extend as scalar functional identities. This is distinct from any question of arithmetic independence of individual special values.

Question 8 (A complete geometric collision calculus). Extend Theorem 5.1 to arbitrary Stieltjes indices and argument-derivative orders, first for three unit-frequency factors. The singular pieces are explicit derivatives of $x^{-1} \log^m x$, so finite partial fractions and order derivatives suggest an algorithm. It should output every power–logarithm term and the finite rate-dependent term, with an integrable uniform remainder. The desired result is an identity for a specified normalization, not a sharper bound.

Question 9 (Hierarchical collisions and composition). Compare the paths $(0, \epsilon, \epsilon + \epsilon^2)$ and $(0, \epsilon, 2\epsilon)$ with iterated merging of a selected pair followed by the third point. Can the finite corrections be organized by nested subsets so that renormalized collision composition is associative? The rate dependence already proved shows why plain constant extraction is insufficient. Specify the intermediate coordinate and subtraction at every stage.

Question 10 (Permutation cancellations beyond total derivatives). The completed generators distinguish global Tornheim data from local endpoint data. Determine which nontrivial permutation or character projections remove the global part for four factors, after the elementary total-derivative identities have been accounted for. A concrete target is total argument order three with equal Stieltjes indices, retaining the complete resonant-subset numerators. Produce either an exact cancellation or a precise surviving multiple-sum kernel.

Question 11 (Finite global kernels for four colliding grids). Theorem 5.2 solves the local completion for any number of factors. Construct a useful global spectral realization for four frequencies on the rank-three lattice $\sum_i p_i n_i = 0$, with a finite sign-region and character decomposition. The first target is $(p_1, p_2, p_3, p_4) = (1, 1, 2, 3)$. Local existence alone should not be presented as a minimal finite reduction of the global multiple Tornheim kernels.

Question 12 (The frozen Gaussian candidates). Retain the exact source vectors for S_6 and the current S_8 . Seek an additional functional or iterated-integral relation beyond the projection mechanisms that already prove S_4 . A successful proof should provide a finite exact row certificate and an analytic justification for every generating row, including its endpoint evaluation. An exclusion from a chosen finite relation space would identify a missing mechanism, not disprove the numerical identity.

6.6 Recommended integration order

The four mathematical sections can be integrated independently after their cited antecedents. Preserve the stated domains, branches, and normalizations, and review the two source edits separately. The accompanying integration notes map labels and source targets while preserving the existing conjecture status.

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